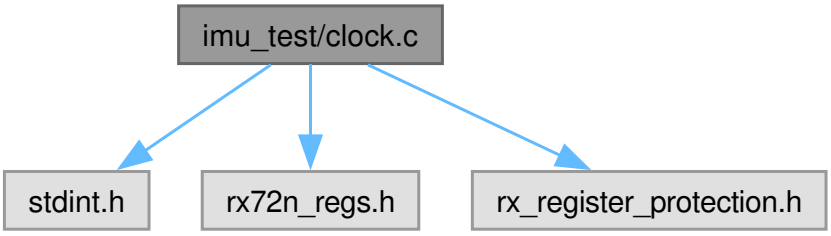
>>

>>

>>

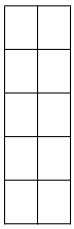
```
#include <stdint.h>
#include "rx72n_regs.h"
#include "rx_register_protection.h"
```



>>

>>>

enum clock_byte_constants_t : uint8_t



```
00058             : uint8_t {
00059   k_sub_clock_stopped = 0x01U, /**< SOSCCR: stop sub-clock oscillator */
00060   k_main_osc_enabled  = 0x00U, /**< MOSCCR: 0 = MOSC running */
00061   k_pll_enabled       = 0x00U, /**< PLLCR2: 0 = PLL running */
00062   k_oscovfsr_moovf    = 0x01U, /**< OSCOVFSR bit 0: MOSC stable */
00063   k_oscovfsr_plovf    = 0x04U, /**< OSCOVFSR bit 2: PLL stable */
00064 } clock_byte_constants_t;
```

enum clock_extra_addrs_t : uintptr_t

```
00027             : uintptr_t {
00028   k_packcr_addr = 0x00080044U, /**< Peripheral clock source: 0=PLL */
00029 } clock_extra_addrs_t;
```

enum clock_sckcr_t : uint32_t

```
00049             : uint32_t {
00050   /*
00051    * SCKCR dividers (matches production k_system_clock_dividers):
00052    *   FCK=/4 (60), ICK=/1 (240), PSTOP1=1, PSTOP0=1, BCK=/4 (60),
00053    *   PCKA=/2 (120), PCKB=/4 (60), PCKC=/4 (60), PCKD=/4 (60).
00054    */
00055   k_sckcr_value = 0x21C21211U,
00056 } clock_sckcr_t;
```

enum clock_timeout_t : uint32_t

```
00066             : uint32_t {
00067   /* Bounded poll counts (NASA Rule 2). We poll the hardware ready flag
00068    * (OSCOVFSR.MOOVF / .PLOVF) so the actual wait is hardware-determined --
00069    * roughly 10 ms physical time regardless of CPU clock. The upper bound
00070    * just prevents an infinite spin if the crystal is missing. ~500k iters
00071    * at LOCO 240 kHz is ~3 s, which is still much shorter than the prior
00072    * fixed 10-second busy-wait. */
00073   k_main_osc_poll_max  = 500000U,
00074   k_pll_lock_poll_max  = 500000U,
00075 } clock_timeout_t;
```

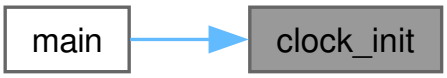
enum clock_word_constants_t : uint16_t

```
00034             : uint16_t {
00035   /*
00036    * PLLCR layout:
00037    *   bits[1:0]  PLIDIV    : 00 = /1
00038    *   bit [4]    PLLSRCSEL : 0 = MOSC, 1 = HOCO
00039    *   bits[13:8] STC       : multiplier = (STC + 1) / 2
00040    *
00041    * MOSC 24 MHz / 1 * 10 = 240 MHz. STC = (10 * 2) - 1 = 19 = 0x13.
00042    * PLLSRCSEL = 0 (MOSC) -> bit 4 clear.
00043    *
00044    * Production rx_clock_power_init.c uses k_pll_multiplier _div  = 0x1300.
00045    */
00046   k_pllcr_mosc_x10 = 0x1300U,
00047 } clock_word_constants_t;
```

void clock_init (
                void)

```
00081 {
00082   volatile rx_system_regs_t *sys = system_regs();
00083
00084   /* Unlock PRC0 (clock) + PRC1 (module stop) write protection. */
00085   *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00086
00087   /* Stop sub-clock oscillator -- not used on STAR. */
00088   sys->sosccr = k_sub_clock_stopped;
00089
00090   /* Start MOSC (24 MHz external crystal). */
00091   sys->mosccr = k_main_osc_enabled;
00092
00093   /* Poll the hardware MOSC-stable flag. Exits as soon as the crystal has
00094    * settled (typically ~10 ms physical time) instead of waiting a fixed
00095    * 2.4M cycles, which on the LOCO 240 kHz boot clock is ~10 seconds. */
00096   for (volatile uint32_t i = 0; i < k_main_osc_poll_max; i++) {
00097     if ((sys->oscovfsr & k_oscovfsr_moovf) != 0U) {
00098       break;
00099     }
00100   }
00101
00102   /* Configure and enable PLL: 24 MHz x 10 / 1 = 240 MHz. */
00103   sys->pllcr  = k_pllcr_mosc_x10;
00104   sys->pllcr2 = k_pll_enabled;
00105
00106   /* Wait for PLL lock (bounded). If we time out, the crystal is missing
00107    * or not driving -- DO NOT switch SCKCR3 to PLL or the chip will halt
00108    * on a dead clock. Stay on LOCO so the LEDs still respond and the user
00109    * can see we got past clock_init. */
00110   bool pll_locked = false;
00111   for (volatile uint32_t i = 0; i < k_pll_lock_poll_max; i++) {
00112     if ((sys->oscovfsr & k_oscovfsr_plovf) != 0U) {
00113       pll_locked = true;
00114       break;
00115     }
00116   }
00117
00118   if (pll_locked) {
00119     /* MANDATORY: raise flash wait state BEFORE switching to >120 MHz. */
00120     *memwait_reg() = k_memwait_one_wait;
00121
00122     /* Apply system clock dividers. */
00123     sys->sckcr = k_sckcr_value;
00124
00125     /* Route UCLK from main PLL (UPLLSEL=0). */
00126     *((volatile uint16_t *)k_packcr_addr) = 0x0000U;
00127
00128     /* Switch system clock source to PLL. */
00129     sys->sckcr3 = k_sckcr3_cksel_pll;
00130   }
00131
00132   /* Re-lock PRCR. */
```

```
00133   *prcr_reg() = k_rx_prcr_lock;
00134 }
```




```
00001 /**
00002  * @file clock.c
00003  * @brief RX72N clock init for the production STAR PCB: MOSC 24 MHz crystal
00004  *        -> PLL 240 MHz -> ICLK=240 / PCKA=120 / PCKB=60 / FCLK=60 MHz.
00005  *
00006  * @details
00007  * Self-contained equivalent of src/rx_clock_power_init.c, stripped of the
00008  * logging, asserts, simulator branches, and PPLL/USB clock paths the motor
00009  * spin test does not need. Configures only the main PLL.
00010  *
00011  * Path: EXTAL 24 MHz -> MOSC -> PLL (x10 / 1) = 240 MHz -> SCKCR=0x21C21211.
00012  * After this routine returns, ICLK is 240 MHz and PCLKA (the GPTW source)
00013  * is 120 MHz, matching k_pclka_hz in libs/rx_hal/inc/rx72n_clock.h.
00014  *
00015  * SPDX-License-Identifier: MIT
00016  * @copyright Copyright (c) 2026 Locked Inc.
00017  */
00018
00019 #include <stdint.h>
00020
00021 #include "rx72n_regs.h"
00022 #include "rx_register_protection.h"
00023
00024 /*
      ============================================================================
00025  * Register addresses not exposed by rx_system_regs_t
00026  *
      ============================================================================
      */
00027 typedef enum : uintptr_t {
00028   k_packcr_addr = 0x00080044U, /**< Peripheral clock source: 0=PLL */
00029 } clock_extra_addrs_t;
00030
00031 /*
      ============================================================================
00032  * Bit / value constants (mirrors src/rx_clock_power_init.c)
00033  *
      ============================================================================
      */
00034 typedef enum : uint16_t {
00035   /*
00036    * PLLCR layout:
00037    *   bits[1:0]  PLIDIV    : 00 = /1
00038    *   bit [4]    PLLSRCSEL : 0 = MOSC, 1 = HOCO
00039    *   bits[13:8] STC       : multiplier = (STC + 1) / 2
00040    *
00041    * MOSC 24 MHz / 1 * 10 = 240 MHz. STC = (10 * 2) - 1 = 19 = 0x13.
00042    * PLLSRCSEL = 0 (MOSC) -> bit 4 clear.
00043    *
00044    * Production rx_clock_power_init.c uses k_pll_multiplier_div  = 0x1300.
00045    */
00046   k_pllcr_mosc_x10 = 0x1300U,
00047 } clock_word_constants_t;
00048
00049 typedef enum : uint32_t {
00050   /*
00051    * SCKCR dividers (matches production k_system_clock_dividers):
00052    *   FCK=/4 (60), ICK=/1 (240), PSTOP1=1, PSTOP0=1, BCK=/4 (60),
00053    *   PCKA=/2 (120), PCKB=/4 (60), PCKC=/4 (60), PCKD=/4 (60).
00054    */
00055   k_sckcr_value = 0x21C21211U,
00056 } clock_sckcr_t;
00057
00058 typedef enum : uint8_t {
00059   k_sub_clock_stopped = 0x01U, /**< SOSCCR: stop sub-clock oscillator */
```

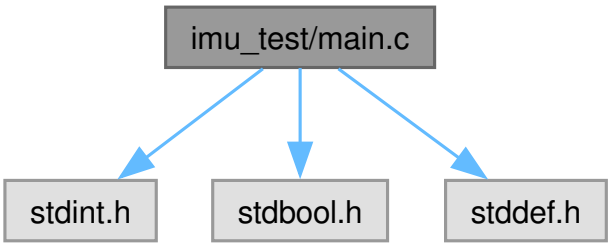
```
00060   k_main_osc_enabled  = 0x00U, /**< MOSCCR: 0 = MOSC running */
00061   k_pll_enabled       = 0x00U, /**< PLLCR2: 0 = PLL running */
00062   k_oscovfsr_moovf    = 0x01U, /**< OSCOVFSR bit 0: MOSC stable */
00063   k_oscovfsr_plovf    = 0x04U, /**< OSCOVFSR bit 2: PLL stable */
00064 } clock_byte_constants_t;
00065
00066 typedef enum : uint32_t {
00067   /* Bounded poll counts (NASA Rule 2). We poll the hardware ready flag
00068    * (OSCOVFSR.MOOVF / .PLOVF) so the actual wait is hardware-determined --
00069    * roughly 10 ms physical time regardless of CPU clock. The upper bound
00070    * just prevents an infinite spin if the crystal is missing. ~500k iters
00071    * at LOCO 240 kHz is ~3 s, which is still much shorter than the prior
00072    * fixed 10-second busy-wait. */
00073   k_main_osc_poll_max  = 500000U,
00074   k_pll_lock_poll_max  = 500000U,
00075 } clock_timeout_t;
00076
00077 /*
      ============================================================================
00078  * clock_init -- bring the chip from POR defaults to PLL 240 / PCKA 120 MHz.
00079  *
      ============================================================================
      */
00080 void clock_init(void)
00081 {
00082   volatile rx_system_regs_t *sys = system_regs();
00083
00084   /* Unlock PRC0 (clock) + PRC1 (module stop) write protection. */
00085   *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00086
00087   /* Stop sub-clock oscillator -- not used on STAR. */
00088   sys->sosccr = k_sub_clock_stopped;
00089
00090   /* Start MOSC (24 MHz external crystal). */
00091   sys->mosccr = k_main_osc_enabled;
00092
00093   /* Poll the hardware MOSC-stable flag. Exits as soon as the crystal has
00094    * settled (typically ~10 ms physical time) instead of waiting a fixed
00095    * 2.4M cycles, which on the LOCO 240 kHz boot clock is ~10 seconds. */
00096   for (volatile uint32_t i = 0; i < k_main_osc_poll_max; i++) {
00097     if ((sys->oscovfsr & k_oscovfsr_moovf) != 0U) {
00098       break;
00099     }
00100   }
00101
00102   /* Configure and enable PLL: 24 MHz x 10 / 1 = 240 MHz. */
00103   sys->pllcr  = k_pllcr_mosc_x10;
00104   sys->pllcr2 = k_pll_enabled;
00105
00106   /* Wait for PLL lock (bounded). If we time out, the crystal is missing
00107    * or not driving -- DO NOT switch SCKCR3 to PLL or the chip will halt
00108    * on a dead clock. Stay on LOCO so the LEDs still respond and the user
00109    * can see we got past clock_init. */
00110   bool pll_locked = false;
00111   for (volatile uint32_t i = 0; i < k_pll_lock_poll_max; i++) {
00112     if ((sys->oscovfsr & k_oscovfsr_plovf) != 0U) {
00113       pll_locked = true;
00114       break;
00115     }
00116   }
00117
00118   if (pll_locked) {
00119     /* MANDATORY: raise flash wait state BEFORE switching to >120 MHz. */
00120     *memwait_reg() = k_memwait_one_wait;
00121
00122     /* Apply system clock dividers. */
00123     sys->sckcr = k_sckcr_value;
00124
00125     /* Route UCLK from main PLL (UPLLSEL=0). */
00126     *((volatile uint16_t *)k_packcr_addr) = 0x0000U;
00127
00128     /* Switch system clock source to PLL. */
00129     sys->sckcr3 = k_sckcr3_cksel_pll;
00130   }
00131
00132   /* Re-lock PRCR. */
00133   *prcr_reg() = k_rx_prcr_lock;
00134 }
```

```
00001 /*
00002  * gpio_test/linker.ld -- Linker script for GPIO breakout board test.
00003  *
```

```
00004  * Same memory map as usb_test: 512 KB internal SRAM, 2 MB internal ROM.
00005  * Includes Renesas-syntax sections (P, B, D, C) needed by ThreadX RXv3 port.
00006  */
00007
00008 MEMORY
00009 {
00010     RAM : ORIGIN = 0x00000004, LENGTH = 0x7FFFC   /* 512 KB - 4 bytes */
00011     ROM : ORIGIN = 0xFFE00000, LENGTH = 0x200000  /* 2 MB             */
00012 }
00013
00014 SECTIONS
00015 {
00016     /* Code -- explicit ROM base so section VMA is anchored correctly. */
00017     .text 0xFFE00000 : AT(0xFFE00000)
00018     {
00019         *(.text*)
00020         *(P)
00021         *(P )
00022         *(.rodata*)
00023         *(C)
00024         *(C )
00025         *(C )
00026         etext = .;
00027     } > ROM
00028
00029     /*
00030      * Relocatable vector table -- startup.S loads INTB with this address.
00031      * Must be 4-byte aligned; only vectors 27 (SWINT) and 28 (CMT0) are
00032      * used, but allocate 256 entries (1024 bytes) for safety.
00033      */
00034     .rvectors ALIGN(4) :
00035     {
00036         __rvectors_start = .;
00037         KEEP(*(.rvectors))
00038         __rvectors_end = .;
00039     } > ROM
00040
00041     /*
00042      * Initialised data -- LMA stays in ROM, VMA in RAM.
00043      * startup.S copies _data..edata from __mdata on boot.
00044      */
00045     __mdata = .;
00046     .data : AT(__mdata)
00047     {
00048         __data = .;
00049         *(.data*)
00050         *(D)
00051         *(D )
00052         *(D )
00053         __edata = .;
00054     } > RAM
00055
00056     /* BSS -- zeroed by startup.S */
00057     .bss :
00058     {
00059         __bss = .;
00060         *(.bss*)
00061         *(COMMON)
00062         *(B)
00063         *(B )
00064         *(B )
00065         __ebss = .;
00066         . = ALIGN(4);
00067         __end = .;
00068     } > RAM AT>RAM
00069
00070     /*
00071      * Stacks -- USP grows down from top of RAM, ISP 4 KB below that.
00072      */
00073     __ustack = ORIGIN(RAM) + LENGTH(RAM);
00074     __istack = __ustack - 0x1000;
00075
00076     /*
00077      * Exception vector table at fixed hardware address 0xFFFFFF80.
00078      * 32 entries -- all unused.
00079      */
00080     .exvectors 0xFFFFFF80 : AT(0xFFFFFF80)
00081     {
00082         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00083         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00084         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00085         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00086         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00087         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00088         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00089         LONG(0xFFFFFFFF); LONG(0xFFFFFFFF); LONG(0xFFFFFFFF);
00090     } > ROM
```

```
00091
00092     /* Fixed reset vector at 0xFFFFFFFC */
00093     .fvectors 0xFFFFFFFC : AT(0xFFFFFFFC)
00094     {
00095         KEEP(*(.fvectors))
00096     } > ROM
00097 }
```

```
#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>
```



*

*

*

*

*

#define I16_LE(
p,
i)

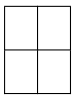
((int16_t)((uint16_t)(p)[(i)] | ((uint16_t)(p)[(i) + 1U] « 8)))

```
#define REG16(
                        a)


 (*(volatile uint16_t *)(uintptr_t)(a))



#define REG32(
                        a)


 (*(volatile uint32_t *)(uintptr_t)(a))



#define REG8(
                        a)


 (*(volatile uint8_t  *)(uintptr_t)(a))
```

```
enum delay_const_t : uint32_t
```

```
00067             : uint32_t {
00068   k_iters_per_ms = 40000U,
00069 } delay_const_t;
```

```
enum gpio_pins_t : uint8_t
```

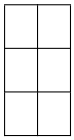


```
00083             : uint8_t {
00084   k_port        = 8,
00085   k_bit_imu_rst  = 3,    /* P83 = active-low BNO055 reset */
00086 } gpio_pins_t;
```

enum poll_max_t : uint32_t

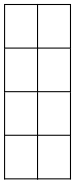
```
00290             : uint32_t {
00291   k_poll_max = 100000U,                  /* ~few ms at -O1 */
00292 } poll_max_t;
```

enum port_base_t : uintptr_t



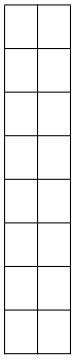
```
00088             : uintptr_t {
00089   k_pdr_base  = 0x0008C000U,
00090   k_podr_base = 0x0008C020U,
00091   k_pmr_base  = 0x0008C060U,
00092 } port_base_t;
```

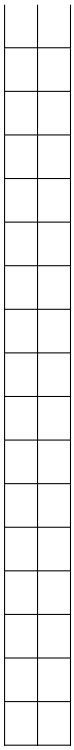
enum riic_addrs_t : uintptr_t



```
00120             : uintptr_t {
00121   k_riic1_base   = 0x00088320U,
00122   k_mstpcrb_addr = 0x00080014U,
00123   k_prcr_addr    = 0x000803FEU,
00124   k_pwpr_addr    = 0x0008C11FU,
00125 } riic_addrs_t;
```

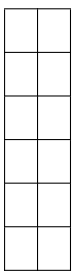
enum riic_bits_t : uint8_t

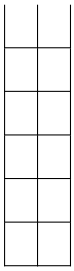




```
00144             : uint8_t {
00145   k_iccr1_iceen   = 0x80,
00146   k_iccr1_iicrst  = 0x40,
00147   k_iccr2_bbsy    = 0x80,
00148   k_iccr2_mst     = 0x40,    /* Controller mode (writable in config phase) */
00149   k_iccr2_trs     = 0x20,    /* Transmit/Receive select (1=transmit) */
00150   k_iccr2_st      = 0x02,
00151   k_iccr2_sp      = 0x08,
00152   k_iccr2_rs      = 0x04,
00153   k_icsr2_al      = 0x02,
00154   k_icsr2_start   = 0x04,
00155   k_icsr2_stop    = 0x08,
00156   k_icsr2_nackf   = 0x10,
00157   k_icsr2_rdrf    = 0x20,
00158   k_icsr2_tend    = 0x40,
00159   k_icsr2_tdre    = 0x80,
00160   k_icmr3_ackbt   = 0x08,
00161   k_icmr3_ackwp   = 0x10,
00162   k_icfer_tmoe    = 0x01,
00163   k_icfer_male    = 0x02,
00164   k_icfer_nale    = 0x04,
00165   k_icfer_sale    = 0x08,
00166   k_icfer_nacke   = 0x10,
00167   k_icfer_nfe     = 0x20,
00168   k_icfer_scle    = 0x40,
00169   k_icfer_fmpe    = 0x80,
00170 } riic_bits_t;
```

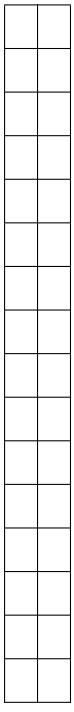
enum riic_off_t : uint8_t





```
00127             : uint8_t {
00128   k_riic_off_iccr1 = 0x00,
00129   k_riic_off_iccr2 = 0x01,
00130   k_riic_off_icmr1 = 0x02,
00131   k_riic_off_icmr3 = 0x04,
00132   k_riic_off_icfer = 0x05,
00133   k_riic_off_icier = 0x07,
00134   k_riic_off_icsr1 = 0x08,
00135   k_riic_off_icsr2 = 0x09,
00136   k_riic_off_icbrl = 0x10,
00137   k_riic_off_icbrh = 0x11,
00138   k_riic_off_icdrt = 0x12,
00139   k_riic_off_icdrr = 0x13,
00140 } riic_off_t;
```

enum sensor_t : uint8_t

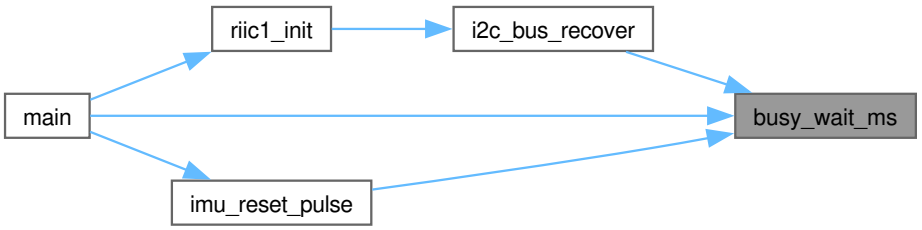


```
00514             : uint8_t {
00515   k_bno055_addr        = 0x28U,
00516   k_bno055_chip_id_reg = 0x00U,
00517   k_bno055_acc_x_lsb   = 0x08U,
00518   k_bno055_grv_x_lsb   = 0x2EU,  /* Gravity vector base, 6 bytes int16 LE, 1 LSB = 1/100 m/s^2 */
00519   k_bno055_lia_x_lsb   = 0x28U,  /* Linear acceleration base (gravity removed by fusion) */
00520   k_bno055_opr_mode    = 0x3DU,  /* Operation mode register */
00521   k_bno055_pwr_mode    = 0x3EU,  /* Power mode register */
00522   k_bno055_sys_trigger = 0x3FU,  /* System trigger (incl. self-test, soft reset) */
00523   k_bno055_unit_sel    = 0x3BU,  /* Output units selection */
00524   k_bno055_page_id     = 0x07U,  /* Register page selector */
00525   k_bno055_calib_stat  = 0x35U,  /* Calibration status (SYS|GYR|ACC|MAG, 2 bits each) */
```

```
00526   k_bno055_mode_config = 0x00U,  /* OPR_MODE: CONFIGMODE */
00527   k_bno055_mode_ndof   = 0x0CU,  /* OPR_MODE: NDOF (full sensor fusion) */
00528   k_bno055_pwr_normal  = 0x00U,  /* PWR_MODE: normal */
00529   k_bmp280_addr        = 0x76U,
00530   k_bmp280_chip_id_reg = 0xD0U,
00531 } sensor_t;
```

void busy_wait_ms (
uint32_t ms) [static]

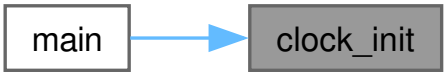
```
00072 {
00073   for (uint32_t m = 0; m < ms; m++) {
00074     for (volatile uint32_t i = 0; i < k_iters_per_ms; i++) {
00075       __asm__ volatile("nop");
00076     }
00077   }
00078 }
```



void clock_init (
void) [extern]

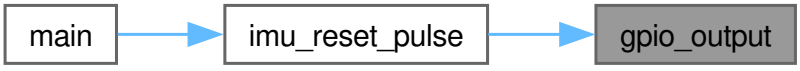
```
00081 {
00082   volatile rx_system_regs_t *sys = system_regs();
00083
00084   /* Unlock PRC0 (clock) + PRC1 (module stop) write protection. */
00085   *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00086
00087   /* Stop sub-clock oscillator -- not used on STAR. */
00088   sys->sosccr = k_sub_clock_stopped;
00089
00090   /* Start MOSC (24 MHz external crystal). */
00091   sys->mosccr = k_main_osc_enabled;
00092
00093   /* Poll the hardware MOSC-stable flag. Exits as soon as the crystal has
00094    * settled (typically ~10 ms physical time) instead of waiting a fixed
00095    * 2.4M cycles, which on the LOCO 240 kHz boot clock is ~10 seconds. */
00096   for (volatile uint32_t i = 0; i < k_main_osc_poll_max; i++) {
00097     if ((sys->oscovfsr & k_oscovfsr_moovf) != 0U) {
00098       break;
00099     }
00100   }
00101
00102   /* Configure and enable PLL: 24 MHz x 10 / 1 = 240 MHz. */
00103   sys->pllcr  = k_pllcr_mosc_x10;
00104   sys->pllcr2 = k_pll_enabled;
00105
00106   /* Wait for PLL lock (bounded). If we time out, the crystal is missing
00107    * or not driving -- DO NOT switch SCKCR3 to PLL or the chip will halt
00108    * on a dead clock. Stay on LOCO so the LEDs still respond and the user
00109    * can see we got past clock_init. */
00110   bool pll_locked = false;
00111   for (volatile uint32_t i = 0; i < k_pll_lock_poll_max; i++) {
00112     if ((sys->oscovfsr & k_oscovfsr_plovf) != 0U) {
```

```
00113       pll_locked = true;
00114       break;
00115     }
00116   }
00117
00118   if (pll_locked) {
00119     /* MANDATORY: raise flash wait state BEFORE switching to >120 MHz. */
00120     *memwait_reg() = k_memwait_one_wait;
00121
00122     /* Apply system clock dividers. */
00123     sys->sckcr = k_sckcr_value;
00124
00125     /* Route UCLK from main PLL (UPLLSEL=0). */
00126     *((volatile uint16_t *)k_packcr_addr) = 0x0000U;
00127
00128     /* Switch system clock source to PLL. */
00129     sys->sckcr3 = k_sckcr3_cksel_pll;
00130   }
00131
00132   /* Re-lock PRCR. */
00133   *prcr_reg() = k_rx_prcr_lock;
00134 }
```



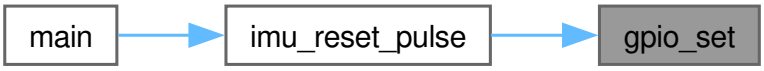
void gpio_output (
uint8_t port,
uint8_t pin) [inline], [static]

```
00095 {
00096   REG8(k_pmr_base + port) &= (uint8_t) ~(1U « pin);
00097   REG8(k_pdr_base + port) |= (uint8_t)(1U « pin);
00098 }
```



void gpio_set (
uint8_t port,
uint8_t pin,
bool high) [inline], [static]

```
00100 {
00101   if (high) { REG8(k_podr_base + port) |= (uint8_t)(1U « pin); }
00102   else      { REG8(k_podr_base + port) &= (uint8_t) ~(1U « pin); }
00103 }
```



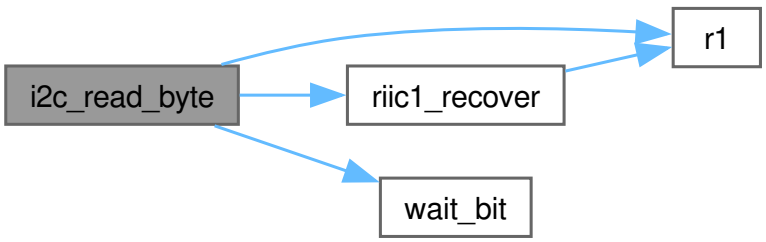
void i2c_bus_recover (
void) [static]

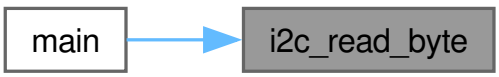
```
00179 {
00180   REG8(0x0008C062U) &= (uint8_t)~((1U « 0) | (1U « 1));   /* PMR: GPIO mode */
00181   REG8(0x0008C084U) |= (uint8_t)((1U « 1) | (1U « 3));    /* P20/P21 NMOS open-drain */
00182   REG8(0x0008C022U) |= (uint8_t)((1U « 0) | (1U « 1));    /* PODR: idle high (released) */
00183   REG8(0x0008C002U) |= (uint8_t)((1U « 0) | (1U « 1));    /* PDR: outputs */
00184
00185   for (uint8_t i = 0; i < 9U; i++) {
00186     REG8(0x0008C022U) &= (uint8_t)~(1U « 1);   /* SCL low */
00187     busy_wait_ms(1);
00188     REG8(0x0008C022U) |= (uint8_t)(1U « 1);    /* SCL high */
00189     busy_wait_ms(1);
00190   }
00191   /* Manual STOP: SDA low while SCL high, then SDA high. */
00192   REG8(0x0008C022U) &= (uint8_t)~(1U « 0);   /* SDA low */
00193   busy_wait_ms(1);
00194   REG8(0x0008C022U) |= (uint8_t)(1U « 0);    /* SDA high (STOP) */
00195   busy_wait_ms(1);
00196
00197   /* Release pins back to input before RIIC1 takes over via PMR=1. RIIC1
00198    * drives the lines on its own once PMR is set; leaving PDR=1 is fine
00199    * per the HW manual but tristating first avoids any glitch. */
00200   REG8(0x0008C002U) &= (uint8_t)~((1U « 0) | (1U « 1));
00201   REG8(0x0008C084U) &= (uint8_t)~((1U « 1) | (1U « 3));   /* clear open-drain */
00202 }
```

bool i2c_read_byte (
uint8_t dev_addr,
uint8_t reg_addr,
uint8_t * out) [static]

```
00310 {
00311   volatile uint8_t *iccr2 = r1(k_riic_off_iccr2);
00312   volatile uint8_t *icsr2 = r1(k_riic_off_icsr2);
00313   volatile uint8_t *icdrt = r1(k_riic_off_icdrt);
00314   volatile uint8_t *icdrr = r1(k_riic_off_icdrr);
00315   volatile uint8_t *icmr3 = r1(k_riic_off_icmr3);
00316
00317   if (!wait_bit(iccr2, k_iccr2_bbsy, false)) {
00318     /* Stuck busy from a previous botched transaction -- nuke the FSM. */
00319     riic1_recover();
00320     if (!wait_bit(iccr2, k_iccr2_bbsy, false)) { return false; }
00321   }
00322   /* Pre-clear any lingering status from the previous read. */
00323   *icsr2 = 0x00U;
00324
00325   /* Controller transmit mode -- mirrors production internal_riic_write_phase().
00326    * Writing MST|TRS while bus is idle is permitted; the bits become read-only
00327    * status during an active transaction. */
00328   *iccr2 = (uint8_t)(k_iccr2_mst | k_iccr2_trs);
```

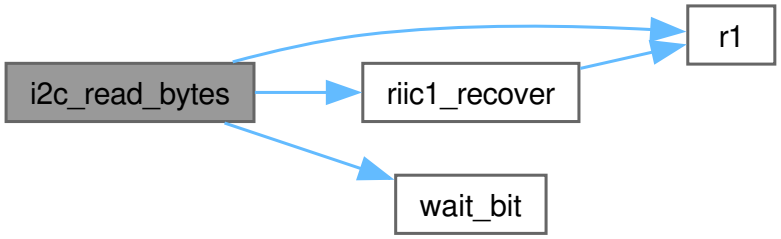
```
00329   *iccr2 |= k_iccr2_st;
00330   if (!wait_bit(icsr2, k_icsr2_start, true)) { s_last_fail_gate = 1U; goto stop_fail; }
00331   *icsr2 &= (uint8_t)~k_icsr2_start;
00332
00333   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_fail_gate = 2U; goto stop_fail; }
00334   *icdrt = (uint8_t)((dev_addr « 1) | 0U);
00335   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_fail_gate = 3U; goto stop_fail; }
00336   if ((*icsr2) & k_icsr2_nackf)             { s_last_fail_gate = 4U; goto stop_fail; }
00337
00338   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_fail_gate = 5U; goto stop_fail; }
00339   *icdrt = reg_addr;
00340   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_fail_gate = 6U; goto stop_fail; }
00341   if ((*icsr2) & k_icsr2_nackf)             { s_last_fail_gate = 7U; goto stop_fail; }
00342
00343   /* Repeated start, then switch to receive mode (TRS=0, MST stays). */
00344   *iccr2 |= k_iccr2_rs;
00345   if (!wait_bit(icsr2, k_icsr2_start, true)) { s_last_fail_gate = 8U; goto stop_fail; }
00346   *icsr2 &= (uint8_t)~k_icsr2_start;
00347   *iccr2 = k_iccr2_mst;
00348
00349   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_fail_gate = 9U; goto stop_fail; }
00350   *icdrt = (uint8_t)((dev_addr « 1) | 1U);
00351   /* Receive-mode address: skip TEND. After ACK, RIIC immediately clocks the
00352    * first data byte and sets RDRF -- TEND never asserts for the addr|R byte
00353    * in controller-receive mode. NACK still surfaces via NACKF if the
00354    * peripheral didn't answer, so we check that before waiting for RDRF. */
00355   if (!wait_bit(icsr2, (uint8_t)(k_icsr2_rdrf | k_icsr2_nackf), true)) {
00356     s_last_fail_gate = 10U;
00357     goto stop_fail;
00358   }
00359   if ((*icsr2) & k_icsr2_nackf)             { s_last_fail_gate = 11U; goto stop_fail; }
00360
00361   *icmr3 |= k_icmr3_ackbt;                  /* NACK after the byte we want */
00362   (void)*icdrr;                             /* dummy read kicks reception */
00363
00364   if (!wait_bit(icsr2, k_icsr2_rdrf, true)) { s_last_fail_gate = 12U; goto stop_fail; }
00365
00366   *icsr2 &= (uint8_t)~k_icsr2_stop;
00367   *iccr2 |= k_iccr2_sp;
00368   *out = *icdrr;
00369
00370   if (!wait_bit(icsr2, k_icsr2_stop, true)) { goto cleanup_only; }
00371 cleanup_only:
00372   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00373   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00374   return (*out != 0xFFU || true);           /* return true regardless of value */
00375
00376 stop_fail:
00377   *iccr2 |= k_iccr2_sp;
00378   (void)wait_bit(icsr2, k_icsr2_stop, true);
00379   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00380   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00381   return false;
00382 }
```





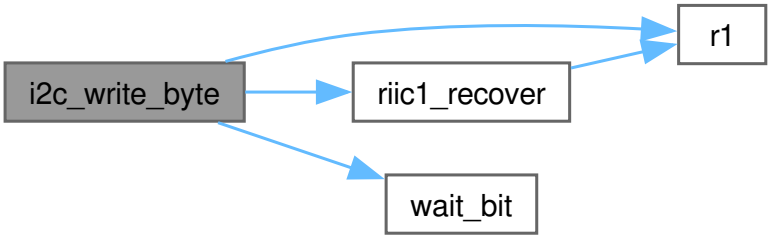
bool i2c_read_bytes (
uint8_t dev_addr,
uint8_t reg_addr,
uint8_t * buf,
uint8_t count) [static]

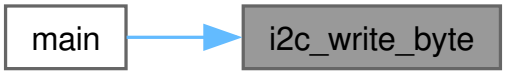
```
00437 {
00438   if (count == 0U) { return false; }
00439
00440   volatile uint8_t *iccr2 = r1(k_riic_off_iccr2);
00441   volatile uint8_t *icsr2 = r1(k_riic_off_icsr2);
00442   volatile uint8_t *icdrt = r1(k_riic_off_icdrt);
00443   volatile uint8_t *icdrr = r1(k_riic_off_icdrr);
00444   volatile uint8_t *icmr3 = r1(k_riic_off_icmr3);
00445
00446   if (!wait_bit(iccr2, k_iccr2_bbsy, false)) {
00447     riic1_recover();
00448     if (!wait_bit(iccr2, k_iccr2_bbsy, false)) { return false; }
00449   }
00450   *icsr2 = 0x00U;
00451
00452   *iccr2 = (uint8_t)(k_iccr2_mst | k_iccr2_trs);
00453   *iccr2 |= k_iccr2_st;
00454   if (!wait_bit(icsr2, k_icsr2_start, true)) { goto rstop_fail; }
00455   *icsr2 &= (uint8_t)~k_icsr2_start;
00456
00457   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { goto rstop_fail; }
00458   *icdrt = (uint8_t)((dev_addr « 1) | 0U);
00459   if (!wait_bit(icsr2, k_icsr2_tend, true)) { goto rstop_fail; }
00460   if ((*icsr2) & k_icsr2_nackf)             { goto rstop_fail; }
00461
00462   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { goto rstop_fail; }
00463   *icdrt = reg_addr;
00464   if (!wait_bit(icsr2, k_icsr2_tend, true)) { goto rstop_fail; }
00465   if ((*icsr2) & k_icsr2_nackf)             { goto rstop_fail; }
00466
00467   *iccr2 |= k_iccr2_rs;
00468   if (!wait_bit(icsr2, k_icsr2_start, true)) { goto rstop_fail; }
00469   *icsr2 &= (uint8_t)~k_icsr2_start;
00470   *iccr2 = k_iccr2_mst;
00471
00472   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { goto rstop_fail; }
00473   *icdrt = (uint8_t)((dev_addr « 1) | 1U);
00474   if (!wait_bit(icsr2, (uint8_t)(k_icsr2_rdrf | k_icsr2_nackf), true)) { goto rstop_fail; }
00475   if ((*icsr2) & k_icsr2_nackf)             { goto rstop_fail; }
00476
00477   /* Dummy read ICDRR to release SCL and start clocking byte 0 into the
00478    * shift register. RX72N HW manual section 38.2.5.3 (controller-receive). */
00479   if (count == 1U) {
00480     /* For a single-byte read, pre-set ACKBT=1 BEFORE the dummy read so the
00481      * byte's ACK slot sends NACK, signalling the peripheral to stop. */
00482     *icmr3 |= k_icmr3_ackbt;
00483   }
00484   (void)*icdrr;
00485
00486   for (uint8_t i = 0; i < count; i++) {
00487     if (!wait_bit(icsr2, k_icsr2_rdrf, true)) { goto rstop_fail; }
00488     if (i == (uint8_t)(count - 2U)) {
00489       /* Next byte is the final one -- set ACKBT=1 so the byte we're about
00490        * to read carries NACK in its 9th-bit slot. */
00491       *icmr3 |= k_icmr3_ackbt;
00492     }
00493     if (i == (uint8_t)(count - 1U)) {
00494       *icsr2 &= (uint8_t)~k_icsr2_stop;
00495       *iccr2 |= k_iccr2_sp;
00496     }
00497     buf[i] = *icdrr;
00498   }
00499
00500   (void)wait_bit(icsr2, k_icsr2_stop, true);
00501   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00502   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00503   return true;
00504
00505 rstop_fail:
00506   *iccr2 |= k_iccr2_sp;
00507   (void)wait_bit(icsr2, k_icsr2_stop, true);
00508   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00509   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00510   return false;
00511 }
```



bool i2c_write_byte (
uint8_t dev_addr,
uint8_t reg_addr,
uint8_t val) [static]

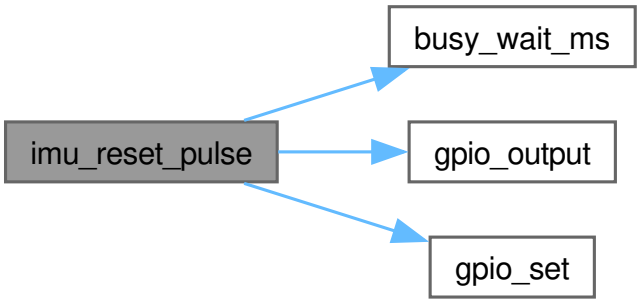
```
00389 {
00390   volatile uint8_t *iccr2 = r1(k_riic_off_iccr2);
00391   volatile uint8_t *icsr2 = r1(k_riic_off_icsr2);
00392   volatile uint8_t *icdrt = r1(k_riic_off_icdrt);
00393
00394   if (!wait_bit(iccr2, k_iccr2_bbsy, false)) {
00395     riic1_recover();
00396     if (!wait_bit(iccr2, k_iccr2_bbsy, false)) { s_last_write_gate = 1U; return false; }
00397   }
00398   *icsr2 = 0x00U;
00399
00400   *iccr2 = (uint8_t)(k_iccr2_mst | k_iccr2_trs);
00401   *iccr2 |= k_iccr2_st;
00402   if (!wait_bit(icsr2, k_icsr2_start, true)) { s_last_write_gate = 2U; goto wstop_fail; }
00403   *icsr2 &= (uint8_t)~k_icsr2_start;
00404
00405   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_write_gate = 3U; goto wstop_fail; }
00406   *icdrt = (uint8_t)((dev_addr « 1) | 0U);
00407   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_write_gate = 4U; goto wstop_fail; }
00408   if ((*icsr2) & k_icsr2_nackf)             { s_last_write_gate = 5U; goto wstop_fail; }
00409
00410   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_write_gate = 6U; goto wstop_fail; }
00411   *icdrt = reg_addr;
00412   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_write_gate = 7U; goto wstop_fail; }
00413   if ((*icsr2) & k_icsr2_nackf)             { s_last_write_gate = 8U; goto wstop_fail; }
00414
00415   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_write_gate = 9U; goto wstop_fail; }
00416   *icdrt = val;
00417   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_write_gate = 10U; goto wstop_fail; }
00418   if ((*icsr2) & k_icsr2_nackf)             { s_last_write_gate = 11U; goto wstop_fail; }
00419
00420   *icsr2 &= (uint8_t)~k_icsr2_stop;
00421   *iccr2 |= k_iccr2_sp;
00422   (void)wait_bit(icsr2, k_icsr2_stop, true);
00423   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00424   return true;
00425
00426 wstop_fail:
00427   *iccr2 |= k_iccr2_sp;
00428   (void)wait_bit(icsr2, k_icsr2_stop, true);
00429   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00430   return false;
00431 }
```





void imu_reset_pulse (
void) [static]

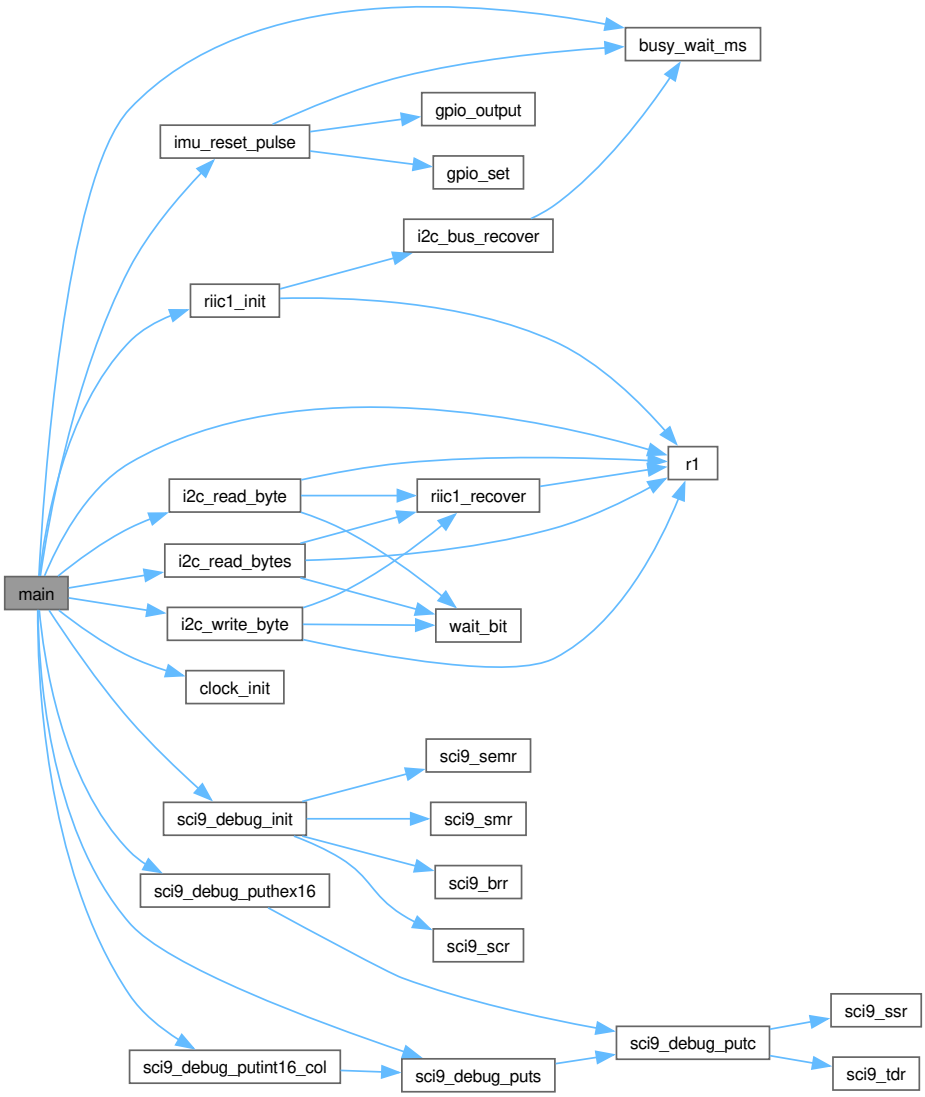
```
00106 {
00107   gpio_output(k_port , k_bit_imu_rst);
00108   gpio_set(k_port , k_bit_imu_rst, true);   /* idle high */
00109   busy_wait_ms(5);
00110   gpio_set(k_port , k_bit_imu_rst, false);  /* assert reset */
00111   busy_wait_ms(5);
00112   gpio_set(k_port , k_bit_imu_rst, true);   /* release */
00113   busy_wait_ms(700);                        /* tBOOT (BNO055 cold start) */
00114 }
```



int main (
                    void)

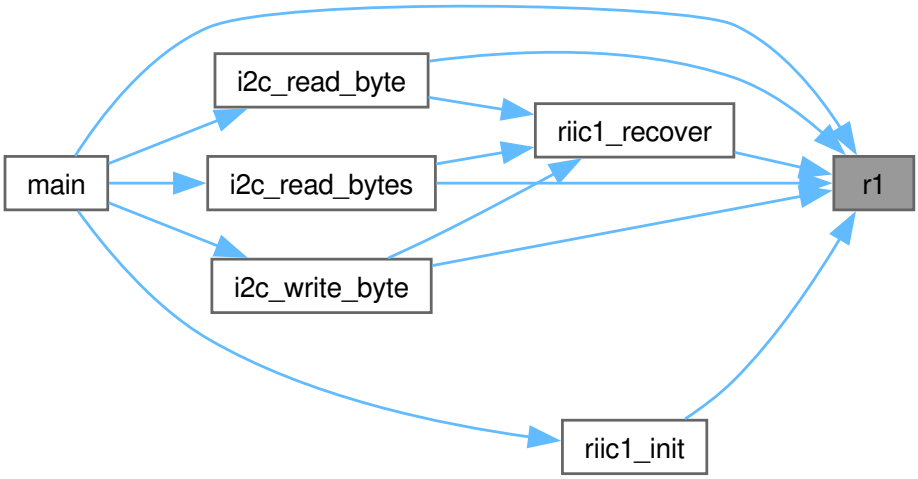
```
00534 {
00535   clock_init();
00536   sci9_debug_init();
00537   /* 5-second window so the operator can attach `cat /dev/ttyACM0` after a
00538    * fresh flash-and-reset and still see step 1..4 messages. */
00539   busy_wait_ms(5000);
00540   sci9_debug_puts("\n=== IMU_TEST start ===\n");
00541
00542   sci9_debug_puts("step 1: imu_reset_pulse (P83 hi-lo-hi + 700 ms)\n");
00543   imu_reset_pulse();
00544   sci9_debug_puts("  done\n");
00545
00546   sci9_debug_puts("step 2: riic1_init\n");
00547   riic1_init();
00548   sci9_debug_puts("  done\n");
00549
00550   sci9_debug_puts("post-init regs:");
00551   sci9_debug_puts(" ICCR1=0x"); sci9_debug_puthex16((uint16_t)*r1(k_riic_off_iccr1));
00552   sci9_debug_puts(" ICCR2=0x"); sci9_debug_puthex16((uint16_t)*r1(k_riic_off_iccr2));
00553   sci9_debug_puts(" ICSR2=0x"); sci9_debug_puthex16((uint16_t)*r1(k_riic_off_icsr2));
00554   sci9_debug_puts("\n");
00555
00556   sci9_debug_puts("step 3: probe BNO055 chip_id @ 0x28 reg 0x00\n");
00557   uint8_t bno_id = 0xFFU;
00558   bool bno_ok = i2c_read_byte(k_bno055_addr, k_bno055_chip_id_reg, &bno_id);
00559   sci9_debug_puts(bno_ok ? "  read ok, chip_id=0x" : "  read FAILED, last=0x");
00560   sci9_debug_puthex16((uint16_t)bno_id);
00561   sci9_debug_puts(bno_id == 0xA0U ? " (BNO055 ok)\n" : " (expected 0xA0)\n");
00562
00563   sci9_debug_puts("step 4: probe BMP280 chip_id @ 0x76 reg 0xD0\n");
00564   uint8_t bmp_id = 0xFFU;
00565   bool bmp_ok = i2c_read_byte(k_bmp280_addr, k_bmp280_chip_id_reg, &bmp_id);
00566   sci9_debug_puts(bmp_ok ? "  read ok, chip_id=0x" : "  read FAILED, last=0x");
00567   sci9_debug_puthex16((uint16_t)bmp_id);
00568   sci9_debug_puts(bmp_id == 0x58U ? " (BMP280 ok)\n" : " (expected 0x58)\n");
00569
00570   sci9_debug_puts("step 5: BNO055 -> NDOF mode (re-init each xact)\n");
00571   riic1_init();
00572   s_last_write_gate = 0U;
00573   bool ok_cfg = i2c_write_byte(k_bno055_addr, k_bno055_opr_mode, k_bno055_mode_config);
00574   uint8_t cfg_gate = s_last_write_gate;
00575   busy_wait_ms(25);
00576   riic1_init();
00577   s_last_write_gate = 0U;
00578   bool ok_ndof = i2c_write_byte(k_bno055_addr, k_bno055_opr_mode, k_bno055_mode_ndof);
00579   uint8_t ndof_gate = s_last_write_gate;
00580   busy_wait_ms(25);
00581   sci9_debug_puts("  cfg="); sci9_debug_puts(ok_cfg ? "ok" : "FAIL");
00582   sci9_debug_puts("@gate=0x"); sci9_debug_puthex16((uint16_t)cfg_gate);
00583   sci9_debug_puts("  ndof="); sci9_debug_puts(ok_ndof ? "ok" : "FAIL");
00584   sci9_debug_puts("@gate=0x"); sci9_debug_puthex16((uint16_t)ndof_gate);
00585   sci9_debug_puts("\n");
00586
00587   sci9_debug_puts("step 6: gravity loop @ 5 Hz (expect |g| ~= 981)\n");
00588   sci9_debug_puts("\n------------------------------------------------------------\n");
00589   sci9_debug_puts("Watching BNO055 @ 2 Hz. All values are raw int16 from the chip.\n");
00590   sci9_debug_puts("Scale factors (per BNO055 datasheet, units = m/s^2, deg/s, deg, uT):\n");
00591   sci9_debug_puts("  ACC / LIA / GRV: divide by 100  (1 LSB = 0.01 m/s^2)\n");
00592   sci9_debug_puts("  GYR            : divide by 16   (1 LSB = 0.0625 deg/s)\n");
00593   sci9_debug_puts("  EUL (hdg/r/p)  : divide by 16   (1 LSB = 0.0625 deg)\n");
00594   sci9_debug_puts("  QUA            : divide by 16384 (1 LSB = 1 / 2^14)\n");
00595   sci9_debug_puts("  MAG            : divide by 16   (1 LSB = 0.0625 uT)\n");
00596   sci9_debug_puts("  temp           : deg C (int8 as-is)\n");
00597   sci9_debug_puts("  cal = SYS«6 | GYR«4 | ACC«2 | MAG (each 0..3)\n");
00598   sci9_debug_puts("------------------------------------------------------------\n");
00599
00600   uint16_t iter = 0;
00601   for (;;) {
00602     /* One burst covers ACC..CALIB_STAT = 46 bytes starting at 0x08. */
00603     uint8_t block[46] = {0};
00604     riic1_init();
00605     bool ok = i2c_read_bytes(k_bno055_addr, k_bno055_acc_x_lsb, block, sizeof(block));
00606
00607     if (!ok) {
00608       sci9_debug_puts("iter=");
00609       sci9_debug_puthex16(iter);
00610       sci9_debug_puts("  READ_FAIL\n");
00611       busy_wait_ms(500);
00612       iter++;
00613       continue;
00614     }
```

```
00615
00616     /* Indexing relative to start-register 0x08:
00617      *   ACC  : 0x08..0x0D -> block[ 0.. 5]
00618      *   MAG  : 0x0E..0x13 -> block[ 6..11]
00619      *   GYR  : 0x14..0x19 -> block[12..17]
00620      *   EUL  : 0x1A..0x1F -> block[18..23]  (heading, roll, pitch)
00621      *   QUA  : 0x20..0x27 -> block[24..31]  (w, x, y, z)
00622      *   LIA  : 0x28..0x2D -> block[32..37]
00623      *   GRV  : 0x2E..0x33 -> block[38..43]
00624      *   TEMP : 0x34       -> block[44]      (int8)
00625      *   CAL  : 0x35       -> block[45]      (status byte)
00626      */
00627 #define I16_LE(p, i) \
00628   ((int16_t)((uint16_t)(p)[(i)] | ((uint16_t)(p)[(i) + 1U] « 8)))
00629
00630     int16_t acc_x = I16_LE(block, 0);  int16_t acc_y = I16_LE(block, 2);  int16_t acc_z = I16_LE(block, 4);
00631     int16_t mag_x = I16_LE(block, 6);  int16_t mag_y = I16_LE(block, 8);  int16_t mag_z = I16_LE(block, 10);
00632     int16_t gyr_x = I16_LE(block, 12); int16_t gyr_y = I16_LE(block, 14); int16_t gyr_z = I16_LE(block, 16);
00633     int16_t eul_h = I16_LE(block, 18); int16_t eul_r = I16_LE(block, 20); int16_t eul_p = I16_LE(block, 22);
00634     int16_t qua_w = I16_LE(block, 24); int16_t qua_x = I16_LE(block, 26);
00635     int16_t qua_y = I16_LE(block, 28); int16_t qua_z = I16_LE(block, 30);
00636     int16_t lia_x = I16_LE(block, 32); int16_t lia_y = I16_LE(block, 34); int16_t lia_z = I16_LE(block, 36);
00637     int16_t grv_x = I16_LE(block, 38); int16_t grv_y = I16_LE(block, 40); int16_t grv_z = I16_LE(block, 42);
00638     int8_t  temp  = (int8_t)block[44];
00639     uint8_t calib = block[45];
00640
00641 #undef I16_LE
00642
00643     sci9_debug_puts("iter=");
00644     sci9_debug_puthex16(iter);
00645     sci9_debug_puts("\n");
00646
00647     sci9_debug_puts("  ACC  x="); sci9_debug_putint16_col(acc_x);
00648     sci9_debug_puts("  y=");      sci9_debug_putint16_col(acc_y);
00649     sci9_debug_puts("  z=");      sci9_debug_putint16_col(acc_z);
00650     sci9_debug_puts("   (raw accel, /100 -> m/s^2)\n");
00651
00652     sci9_debug_puts("  MAG  x="); sci9_debug_putint16_col(mag_x);
00653     sci9_debug_puts("  y=");      sci9_debug_putint16_col(mag_y);
00654     sci9_debug_puts("  z=");      sci9_debug_putint16_col(mag_z);
00655     sci9_debug_puts("   (magnetometer, /16 -> uT)\n");
00656
00657     sci9_debug_puts("  GYR  x="); sci9_debug_putint16_col(gyr_x);
00658     sci9_debug_puts("  y=");      sci9_debug_putint16_col(gyr_y);
00659     sci9_debug_puts("  z=");      sci9_debug_putint16_col(gyr_z);
00660     sci9_debug_puts("   (gyroscope, /16 -> deg/s)\n");
00661
00662     sci9_debug_puts("  EUL  h="); sci9_debug_putint16_col(eul_h);
00663     sci9_debug_puts("  r=");      sci9_debug_putint16_col(eul_r);
00664     sci9_debug_puts("  p=");      sci9_debug_putint16_col(eul_p);
00665     sci9_debug_puts("   (euler heading/roll/pitch, /16 -> deg)\n");
00666
00667     sci9_debug_puts("  QUA  w="); sci9_debug_putint16_col(qua_w);
00668     sci9_debug_puts("  x=");      sci9_debug_putint16_col(qua_x);
00669     sci9_debug_puts("  y=");      sci9_debug_putint16_col(qua_y);
00670     sci9_debug_puts("  z=");      sci9_debug_putint16_col(qua_z);
00671     sci9_debug_puts("  (quaternion, /16384)\n");
00672
00673     sci9_debug_puts("  LIA  x="); sci9_debug_putint16_col(lia_x);
00674     sci9_debug_puts("  y=");      sci9_debug_putint16_col(lia_y);
00675     sci9_debug_puts("  z=");      sci9_debug_putint16_col(lia_z);
00676     sci9_debug_puts("   (linear accel, gravity removed, /100)\n");
00677
00678     sci9_debug_puts("  GRV  x="); sci9_debug_putint16_col(grv_x);
00679     sci9_debug_puts("  y=");      sci9_debug_putint16_col(grv_y);
00680     sci9_debug_puts("  z=");      sci9_debug_putint16_col(grv_z);
00681     sci9_debug_puts("   (gravity vector, /100)\n");
00682
00683     sci9_debug_puts("  temp="); sci9_debug_putint16_col((int16_t)temp);
00684     sci9_debug_puts(" degC     cal=0x"); sci9_debug_puthex16((uint16_t)calib);
00685     sci9_debug_puts(" (SYS="); sci9_debug_putint16_col((int16_t)((calib » 6) & 0x3U));
00686     sci9_debug_puts(" GYR="); sci9_debug_putint16_col((int16_t)((calib » 4) & 0x3U));
00687     sci9_debug_puts(" ACC="); sci9_debug_putint16_col((int16_t)((calib » 2) & 0x3U));
00688     sci9_debug_puts(" MAG="); sci9_debug_putint16_col((int16_t)(calib & 0x3U));
00689     sci9_debug_puts(")\n\n");
00690
00691     busy_wait_ms(500);
00692     iter++;
00693   }
00694 }
```



volatile uint8_t * r1 (
uint8_t off) [inline], [static]

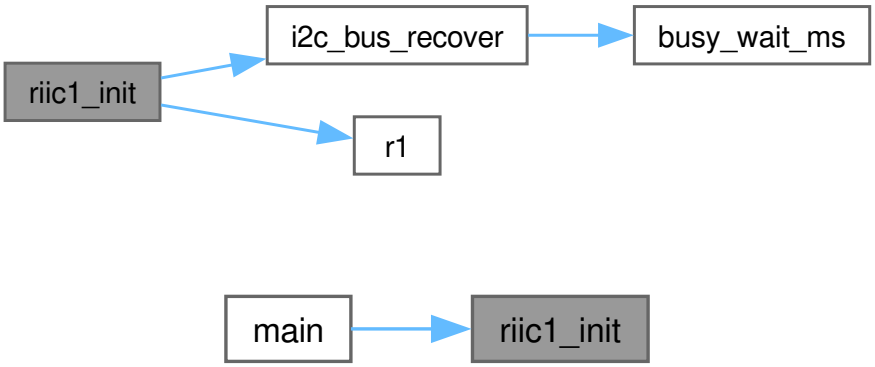
00142 { return &REG8(k_riic1_base + off); }



void riic1_init (
void) [static]

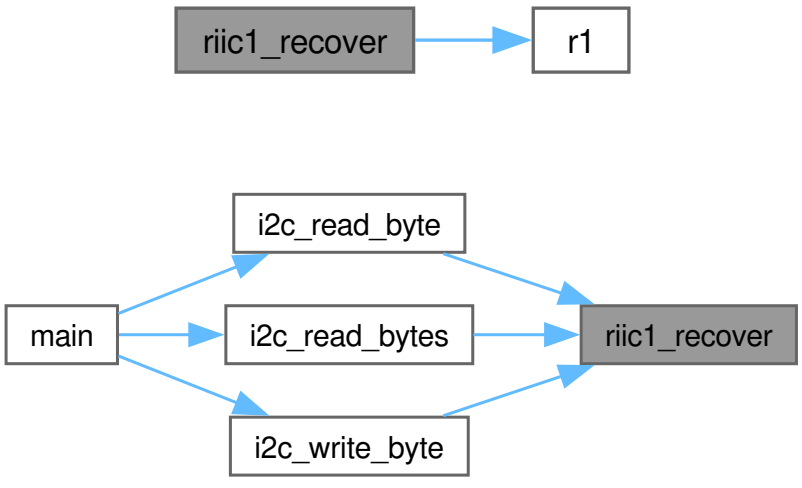
```
00205 {
00206  /* Step 0: 9-clock SCL bit-bang to flush a stuck peripheral that may be
00207   * holding SDA low from an aborted previous transaction. Runs while pins
00208   * are still GPIO (PMR=0) so we can drive them directly. */
00209  i2c_bus_recover();
00210
00211  /* Step 1: clear MSTPCRB.MSTPB20 (RIIC1) under PRCR unlock 0xA50B
00212   * (PRC1+PRC3; 0xA50F asserts a reserved bit per the cross-check fix). */
00213  *(volatile uint16_t *)k_prcr_addr  = 0xA50BU;
00214  *(volatile uint32_t *)k_mstpcrb_addr &= ~(uint32_t)(1UL « 20);
00215  *(volatile uint16_t *)k_prcr_addr  = 0xA500U;
00216
00217  /* Step 2: PFS for RIIC1 SDA1=P20, SCL1=P21. PSEL = 0x0F per RX72N
00218   * HW manual chapter 23 Table 23.6. */
00219  *(volatile uint8_t *)k_pwpr_addr = 0x00U;
00220  *(volatile uint8_t *)k_pwpr_addr = 0x40U;     /* unlock PFS */
00221  REG8(0x0008C140U + 2U * 8U + 0U) = 0x0FU;     /* P20 SDA1 */
00222  REG8(0x0008C140U + 2U * 8U + 1U) = 0x0FU;     /* P21 SCL1 */
00223  *(volatile uint8_t *)k_pwpr_addr = 0x00U;
00224  *(volatile uint8_t *)k_pwpr_addr = 0x80U;     /* lock PFS */
00225
00226  /* Step 3: PMR=1 to route P20/P21 to RIIC1. */
00227  REG8(0x0008C062U) |= (uint8_t)((1U « 0) | (1U « 1));
00228
00229  /* Step 4: per production riic.c: pulse internal reset (IICRST=1 then
00230   * IICRST=0) so the RIIC enters configuration state with both IICRST
00231   * and ICEEN cleared. Configuration registers are writable in this
00232   * state. Setting ICBRH/ICBRL while IICRST=1 silently no-ops -- we
00233   * already saw that empirically. */
00234  *r1(k_riic_off_iccr1) = k_iccr1_iicrst;        /* assert internal reset */
00235  *r1(k_riic_off_iccr1) = 0x00U;                 /* clear all bits (IICRST=0) */
00236
00237  /* Step 5: bit rate. RX72N HW manual Table 38.x: lower 5 bits hold
00238   * the BRL/BRH count; upper 3 bits are reserved (read-as-1, write
00239   * value ignored). Writing the raw 5-bit value is fine. Production
00240   * uses internal_calculate_bit_rate(); we hard-code values that give
00241   * ~100 kHz at PCLKB=60 MHz (or scaled if PCLKB is actually 120). */
00242  *r1(k_riic_off_icbrh) = 23U;
00243  *r1(k_riic_off_icbrl) = 27U;
00244
00245  /* Step 6: mode/timing. Mirror production riic_init: only program ICMR1
00246   * (controller-mode, 7-bit addressing); leave ICMR2/ICMR3 at reset (0x00)
00247   * and DO NOT touch ICFER -- its reset value 0x72 enables the on-chip
00248   * noise filter (NFE) and clock-sync (SCLE) which the peripheral relies
00249   * on. Overriding it with our own bitmask clobbers NFE and silently
00250   * breaks edge detection. */
00251  *r1(k_riic_off_icmr1) = 0x08U;
00252
00253  /* Step 7: enable peripheral. */
00254  *r1(k_riic_off_iccr1) = k_iccr1_iceen;         /* ICEEN=1, IICRST=0 */
00255
00256  /* Step 8: kick RIIC out of post-enable "stuck low" by clearing SOWP and
00257   * forcing SCLO/SDAO high (open-drain release), then re-locking SOWP. The
00258   * peripheral comes out of IICRST with SCLO=SDAO=0 (driving low) and only
00259   * releases on the first STOP it sees. We synthesize that release by hand. */
```

```
00260   uint8_t iccr1 = *r1(k_riic_off_iccr1);
00261   iccr1 &= (uint8_t)~(1U « 4);                  /* clear SOWP -> writable */
00262   iccr1 |= (uint8_t)((1U « 3) | (1U « 2));     /* SCLO=1, SDAO=1 (release) */
00263   *r1(k_riic_off_iccr1) = iccr1;
00264   iccr1 |= (uint8_t)(1U « 4);                  /* re-lock SOWP */
00265   *r1(k_riic_off_iccr1) = iccr1;
00266 }
```



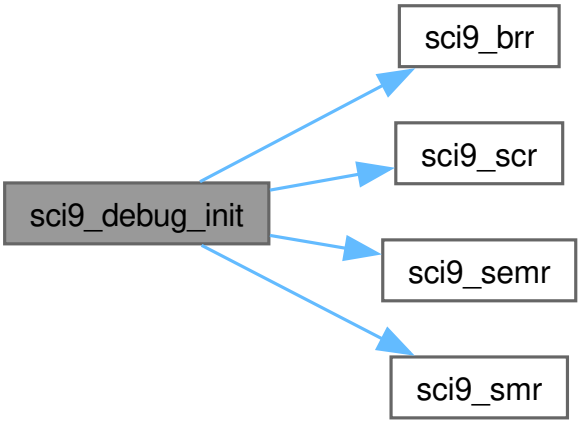
void riic1_recover (
void) [static]

```
00272 {
00273   *r1(k_riic_off_iccr1) = k_iccr1_iicrst;       /* assert IICRST (ICEEN=0) */
00274   *r1(k_riic_off_iccr1) = 0x00U;                /* deassert, configuration state */
00275   /* IICRST cycle wipes ICBRH/L/ICMRx/ICFER back to reset (0xFF). Restore. */
00276   *r1(k_riic_off_icbrh) = 23U;
00277   *r1(k_riic_off_icbrl) = 27U;
00278   *r1(k_riic_off_icmr1) = 0x08U;
00279   *r1(k_riic_off_icmr3) = k_icmr3_ackwp;
00280   /* Drop k_icfer_tmoe -- timeout monitoring leaves RIIC stuck if SCL
00281    * doesn't toggle. Rely on our software wait_bit() bound instead. */
00282   *r1(k_riic_off_icfer) = (uint8_t)(k_icfer_male | k_icfer_nale |
00283                                     k_icfer_sale | k_icfer_nacke | k_icfer_scle);
00284   *r1(k_riic_off_icsr2) = 0x00U;                /* clear all status flags */
00285   *r1(k_riic_off_iccr1) = k_iccr1_iceen;        /* re-enable */
00286 }
```



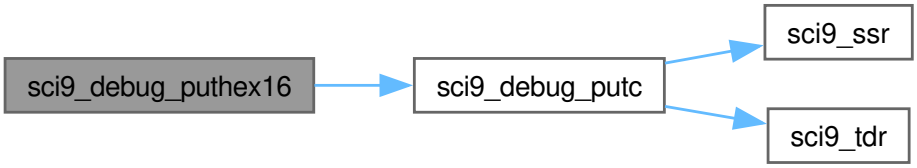
void sci9_debug_init (
void) [extern]

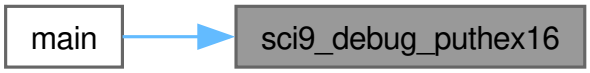
```
00105 {
00106     /* Step 1: release SCI9 from module stop. SCI9 = MSTPCRC bit 26 per
00107      * RX72N HW manual page 410 (NOT MSTPCRB.bit22 -- that's PDC). */
00108     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00109     system_regs()->mstpcrc &= ~(uint32_t)(1UL « (uint32_t)k_mstpc_sci9_bit);
00110     *prcr_reg() = k_rx_prcr_lock;
00111
00112     /* Step 2: configure PB6 (RXD9) and PB7 (TXD9) via MPC.  Use the struct
00113      * accessor so the compiler computes the correct PFS offsets via
00114      * static_assert-checked layout. */
00115     mpc()->pwpr   = k_pwpr_unlock_b0; /* clear B0WI */
00116     mpc()->pwpr   = k_pwpr_pfswe;     /* allow PFS writes */
00117     mpc()->pb6pfs = k_psel_sci9;
00118     mpc()->pb7pfs = k_psel_sci9;
00119     mpc()->pwpr   = k_pwpr_unlock_b0; /* clear PFSWE */
00120     mpc()->pwpr   = k_pwpr_b0wi;      /* re-protect */
00121
00122     /* Step 3: PDR before PMR (production uart.c does this).  PB7 = output
00123      * for TX, PB6 = input for RX.  Then flip PMR to peripheral mode. */
00124     portb()->pdr |= (uint8_t)k_pb7_mask;
00125     portb()->pdr &= (uint8_t) ~k_pb6_mask;
00126     portb()->pmr |= (uint8_t)(k_pb6_mask | k_pb7_mask);
00127
00128     /* Step 4: SCI9 init.  Match the production order:
00129      *   SCR=0 -> SMR -> BRR -> settle -> SCR=TE|RE -> SEMR.
00130      * Don't touch SCMR (reset default 0xF2 is correct). */
00131     *sci9_scr() = k_scr_disabled;
00132     *sci9_smr() = k_smr n1_async;
00133     *sci9_brr() = k_brr ;
00134
00135     /* Wait roughly one bit time before enabling TX/RX. */
00136     for (volatile uint32_t i = 0U; i < (uint32_t)k_brr_settle_loops; i++) {
00137         ___asm___ volatile("nop");
00138     }
00139
00140     *sci9_scr()  = k_scr_txrx_enabled;
00141     *sci9_semr() = k_semr_default;
00142 }
```



void sci9_debug_puthex16 (
uint16_t v) [extern]

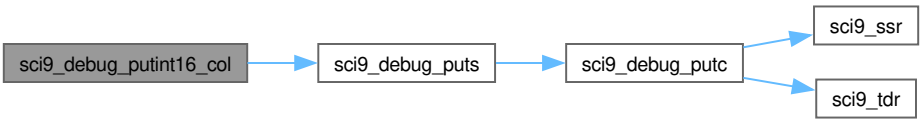
```
00180 {
00181     static const char hex_digits[] = "0123456789ABCDEF";
00182     sci9_debug_putc('0');
00183     sci9_debug_putc('x');
00184     for (int8_t shift = (int8_t)k_hex_shift_init16; shift >= 0;
00185          shift -= (int8_t)k_hex_shift_step) {
00186         sci9_debug_putc(hex_digits[(v » (uint8_t)shift) & (uint8_t)k_hex_nibble_mask]);
00187     }
00188 }
```





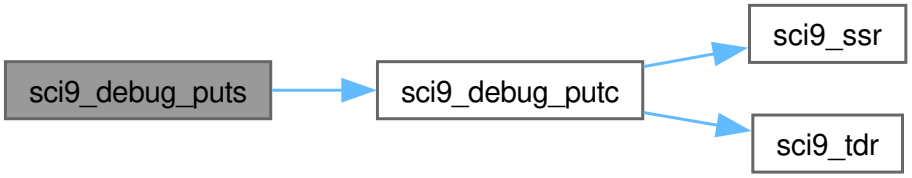
void sci9_debug_putint16_col (
int16_t v) [static]

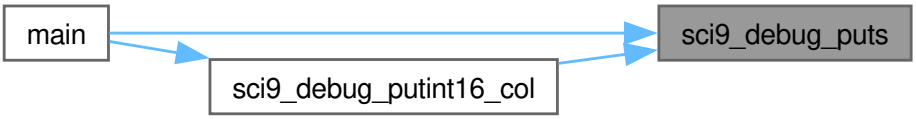
```
00035 {
00036   char     tmp[8] = {0};
00037   uint16_t mag    = (v < 0) ? (uint16_t)(-((int32_t)v)) : (uint16_t)v;
00038   uint8_t  n      = 0;
00039   if (mag == 0U) {
00040     tmp[n++] = '0';
00041   } else {
00042     while (mag > 0U && n < (uint8_t)(sizeof(tmp) - 1U)) {
00043       tmp[n++] = (char)('0' + (mag % 10U));
00044       mag /= 10U;
00045     }
00046   }
00047   char out[9] = {0};
00048   uint8_t idx = 0;
00049   const uint8_t total = (uint8_t)(n + (v < 0 ? 1U : 0U));
00050   for (uint8_t p = total; p < 7U; p++) {
00051     out[idx++] = ' ';
00052   }
00053   if (v < 0) {
00054     out[idx++] = '-';
00055   }
00056   while (n > 0U) {
00057     out[idx++] = tmp[--n];
00058   }
00059   out[idx] = '\0';
00060   sci9_debug_puts(out);
00061 }
```



void sci9_debug_puts (
const char * s) [extern]

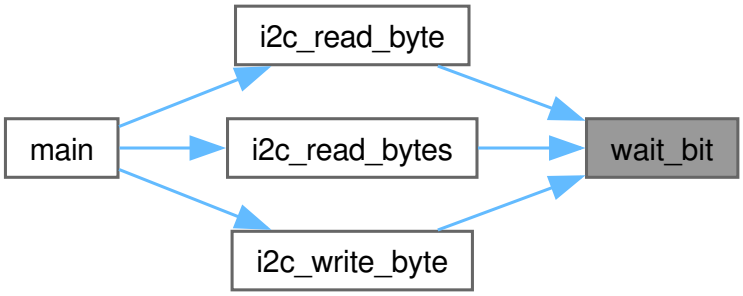
```
00156 {
00157     if (s == (const char *)0) {
00158         return;
00159     }
00160     while (*s != '\0') {
00161         if (*s == '\n') {
00162             sci9_debug_putc('\r');
00163         }
00164         sci9_debug_putc(*s++);
00165     }
00166 }
```





bool wait_bit (
volatile uint8_t * reg,
uint8_t mask,
bool want_set) [static]

```
00295 {
00296   for (uint32_t i = 0; i < k_poll_max; i++) {
00297     bool now = ((*reg) & mask) != 0U;
00298     if (now == want_set) { return true; }
00299   }
00300   return false;
00301 }
```



volatile uint8_t s_last_fail_gate = 0U [static]

volatile uint8_t s_last_write_gate = 0U [static]

```
00001 /**
00002  * @file main.c
00003  * @brief BNO055 + BMP280 over RIIC1 smoke test on the production STAR PCB.
00004  *
00005  * Bare-metal direct-register I2C probe -- prove the I2C wiring, RIIC1
00006  * peripheral config, BNO055 reset+boot, and BMP280 are all healthy.
00007  * Each step prints over SCI9 BEFORE running so a hang is visible.
00008  *
00009  * Wiring (from src/inc/hardware_config.h):
00010  *   IMU_SCL  = P21 / SCL1  pin 36   RIIC1 clock
00011  *   IMU_SDA  = P20 / SDA1  pin 37   RIIC1 data
00012  *   IMU_RST  = P83         pin 58   active-low BNO055 reset
00013  *
00014  * Devices on RIIC1:
00015  *   BNO055  @ 0x28 -- expects chip_id = 0xA0 at reg 0x00
00016  *   BMP280  @ 0x76 -- expects chip_id = 0x58 at reg 0xD0
00017  *
00018  * SPDX-License-Identifier: MIT
00019  * @copyright Copyright (c) 2026 Locked Inc.
00020  */
```

```
00021
00022 #include <stdint.h>
00023 #include <stdbool.h>
00024 #include <stddef.h>
00025
00026 extern void clock_init(void);
00027 extern void sci9_debug_init(void);
00028 extern void sci9_debug_puts(const char *s);
00029 extern void sci9_debug_puthex16(uint16_t v);
00030
00031 /* Signed-decimal printer for int16 values, right-aligned in 7-char column.
00032  * Builds the string in a local buffer first so sign, digits, and padding
00033  * are emitted in one correct order. */
00034 static void sci9_debug_putint16_col(int16_t v)
00035 {
00036   char     tmp[8] = {0};
00037   uint16_t mag    = (v < 0) ? (uint16_t)(-((int32_t)v)) : (uint16_t)v;
00038   uint8_t  n      = 0;
00039   if (mag == 0U) {
00040     tmp[n++] = '0';
00041   } else {
00042     while (mag > 0U && n < (uint8_t)(sizeof(tmp) - 1U)) {
00043       tmp[n++] = (char)('0' + (mag % 10U));
00044       mag /= 10U;
00045     }
00046   }
00047   char out[9] = {0};
00048   uint8_t idx = 0;
00049   const uint8_t total = (uint8_t)(n + (v < 0 ? 1U : 0U));
00050   for (uint8_t p = total; p < 7U; p++) {
00051     out[idx++] = ' ';
00052   }
00053   if (v < 0) {
00054     out[idx++] = '-';
00055   }
00056   while (n > 0U) {
00057     out[idx++] = tmp[--n];
00058   }
00059   out[idx] = '\0';
00060   sci9_debug_puts(out);
00061 }
00062
00063 #define REG8(a)  (*(volatile uint8_t  *)(uintptr_t)(a))
00064 #define REG16(a) (*(volatile uint16_t *)(uintptr_t)(a))
00065 #define REG32(a) (*(volatile uint32_t *)(uintptr_t)(a))
00066
00067 typedef enum : uint32_t {
00068   k_iters_per_ms = 40000U,
00069 } delay_const_t;
00070
00071 static void busy_wait_ms(uint32_t ms)
00072 {
00073   for (uint32_t m = 0; m < ms; m++) {
00074     for (volatile uint32_t i = 0; i < k_iters_per_ms; i++) {
00075       __asm__ volatile("nop");
00076     }
00077   }
00078 }
00079
00080 /* ---------------------------------------------------------------- */
00081 /* GPIO helpers (LEDs + IMU reset on P83) */
00082 /* ---------------------------------------------------------------- */
00083 typedef enum : uint8_t {
00084   k_port        = 8,
00085   k_bit_imu_rst  = 3,    /* P83 = active-low BNO055 reset */
00086 } gpio_pins_t;
00087
00088 typedef enum : uintptr_t {
00089   k_pdr_base  = 0x0008C000U,
00090   k_podr_base = 0x0008C020U,
00091   k_pmr_base  = 0x0008C060U,
00092 } port_base_t;
00093
00094 static inline void gpio_output(uint8_t port, uint8_t pin)
00095 {
00096   REG8(k_pmr_base + port) &= (uint8_t) ~(1U << pin);
00097   REG8(k_pdr_base + port) |= (uint8_t)(1U << pin);
00098 }
00099 static inline void gpio_set(uint8_t port, uint8_t pin, bool high)
00100 {
00101   if (high) { REG8(k_podr_base + port) |= (uint8_t)(1U << pin); }
00102   else      { REG8(k_podr_base + port) &= (uint8_t) ~(1U << pin); }
00103 }
00104
00105 static void imu_reset_pulse(void)
00106 {
00107   gpio_output(k_port , k_bit_imu_rst);
```

```
00108   gpio_set(k_port , k_bit_imu_rst, true);   /* idle high */
00109   busy_wait_ms(5);
00110   gpio_set(k_port , k_bit_imu_rst, false);  /* assert reset */
00111   busy_wait_ms(5);
00112   gpio_set(k_port , k_bit_imu_rst, true);   /* release */
00113   busy_wait_ms(700);                        /* tBOOT (BNO055 cold start) */
00114 }
00115
00116 /* ---------------------------------------------------------------------- */
00117 /* RIIC1 direct-register driver. RIIC1 base = 0x00088321 per RX72N HW   */
00118 /* manual chapter 38. Module-stop bit = MSTPCRB.MSTPB20.                */
00119 /* ---------------------------------------------------------------------- */
00120 typedef enum : uintptr_t {
00121   k_riic1_base   = 0x00088320U,
00122   k_mstpcrb_addr = 0x00080014U,
00123   k_prcr_addr    = 0x000803FEU,
00124   k_pwpr_addr    = 0x0008C11FU,
00125 } riic_addrs_t;
00126
00127 typedef enum : uint8_t {
00128   k_riic_off_iccr1 = 0x00,
00129   k_riic_off_iccr2 = 0x01,
00130   k_riic_off_icmr1 = 0x02,
00131   k_riic_off_icmr3 = 0x04,
00132   k_riic_off_icfer = 0x05,
00133   k_riic_off_icier = 0x07,
00134   k_riic_off_icsr1 = 0x08,
00135   k_riic_off_icsr2 = 0x09,
00136   k_riic_off_icbrl = 0x10,
00137   k_riic_off_icbrh = 0x11,
00138   k_riic_off_icdrt = 0x12,
00139   k_riic_off_icdrr = 0x13,
00140 } riic_off_t;
00141
00142 static inline volatile uint8_t *r1(uint8_t off) { return &REG8(k_riic1_base + off); }
00143
00144 typedef enum : uint8_t {
00145   k_iccr1_iceen   = 0x80,
00146   k_iccr1_iicrst  = 0x40,
00147   k_iccr2_bbsy    = 0x80,
00148   k_iccr2_mst     = 0x40,   /* Controller mode (writable in config phase) */
00149   k_iccr2_trs     = 0x20,   /* Transmit/Receive select (1=transmit) */
00150   k_iccr2_st      = 0x02,
00151   k_iccr2_sp      = 0x08,
00152   k_iccr2_rs      = 0x04,
00153   k_icsr2_al      = 0x02,
00154   k_icsr2_start   = 0x04,
00155   k_icsr2_stop    = 0x08,
00156   k_icsr2_nackf   = 0x10,
00157   k_icsr2_rdrf    = 0x20,
00158   k_icsr2_tend    = 0x40,
00159   k_icsr2_tdre    = 0x80,
00160   k_icmr3_ackbt   = 0x08,
00161   k_icmr3_ackwp   = 0x10,
00162   k_icfer_tmoe    = 0x01,
00163   k_icfer_male    = 0x02,
00164   k_icfer_nale    = 0x04,
00165   k_icfer_sale    = 0x08,
00166   k_icfer_nacke   = 0x10,
00167   k_icfer_nfe     = 0x20,
00168   k_icfer_scle    = 0x40,
00169   k_icfer_fmpe    = 0x80,
00170 } riic_bits_t;
00171
00172 /* Bit-bang SCL 9 times to flush any partial-byte a stuck I2C peripheral may
00173  * be holding SDA low for, then synthesize a STOP. Standard I2C bus-recovery
00174  * dance per NXP UM10204 section 3.1.16. Runs BEFORE we hand the pins to the
00175  * RIIC peripheral. ODR encodes pin drive: 2 bits per pin within ODR0 (pins
00176  * 0-3); the LSB of each pair is reserved, the MSB selects N-ch open-drain.
00177  * For P20: bit 1 of ODR0. For P21: bit 3 of ODR0. */
00178 static void i2c_bus_recover(void)
00179 {
00180   REG8(0x0008C062U) &= (uint8_t)~((1U << 0) | (1U << 1));   /* PMR: GPIO mode */
00181   REG8(0x0008C084U) |= (uint8_t)((1U << 1) | (1U << 3));    /* P20/P21 NMOS open-drain */
00182   REG8(0x0008C022U) |= (uint8_t)((1U << 0) | (1U << 1));    /* PODR: idle high (released) */
00183   REG8(0x0008C002U) |= (uint8_t)((1U << 0) | (1U << 1));    /* PDR: outputs */
00184
00185   for (uint8_t i = 0; i < 9U; i++) {
00186     REG8(0x0008C022U) &= (uint8_t)~(1U << 1);   /* SCL low */
00187     busy_wait_ms(1);
00188     REG8(0x0008C022U) |= (uint8_t)(1U << 1);    /* SCL high */
00189     busy_wait_ms(1);
00190   }
00191   /* Manual STOP: SDA low while SCL high, then SDA high. */
00192   REG8(0x0008C022U) &= (uint8_t)~(1U << 0);   /* SDA low */
00193   busy_wait_ms(1);
00194   REG8(0x0008C022U) |= (uint8_t)(1U << 0);    /* SDA high (STOP) */
```

```
00195   busy_wait_ms(1);
00196
00197   /* Release pins back to input before RIIC1 takes over via PMR=1. RIIC1
00198    * drives the lines on its own once PMR is set; leaving PDR=1 is fine
00199    * per the HW manual but tristating first avoids any glitch. */
00200   REG8(0x0008C002U) &= (uint8_t)~((1U << 0) | (1U << 1));
00201   REG8(0x0008C084U) &= (uint8_t)~((1U << 1) | (1U << 3));   /* clear open-drain */
00202 }
00203
00204 static void riic1_init(void)
00205 {
00206   /* Step 0: 9-clock SCL bit-bang to flush a stuck peripheral that may be
00207    * holding SDA low from an aborted previous transaction. Runs while pins
00208    * are still GPIO (PMR=0) so we can drive them directly. */
00209   i2c_bus_recover();
00210
00211   /* Step 1: clear MSTPCRB.MSTPB20 (RIIC1) under PRCR unlock 0xA50B
00212    * (PRC1+PRC3; 0xA50F asserts a reserved bit per the cross-check fix). */
00213   *(volatile uint16_t *)k_prcr_addr  = 0xA50BU;
00214   *(volatile uint32_t *)k_mstpcrb_addr &= ~(uint32_t)(1UL << 20);
00215   *(volatile uint16_t *)k_prcr_addr  = 0xA500U;
00216
00217   /* Step 2: PFS for RIIC1 SDA1=P20, SCL1=P21. PSEL = 0x0F per RX72N
00218    * HW manual chapter 23 Table 23.6. */
00219   *(volatile uint8_t *)k_pwpr_addr = 0x00U;
00220   *(volatile uint8_t *)k_pwpr_addr = 0x40U;     /* unlock PFS */
00221   REG8(0x0008C140U + 2U * 8U + 0U) = 0x0FU;     /* P20 SDA1 */
00222   REG8(0x0008C140U + 2U * 8U + 1U) = 0x0FU;     /* P21 SCL1 */
00223   *(volatile uint8_t *)k_pwpr_addr = 0x00U;
00224   *(volatile uint8_t *)k_pwpr_addr = 0x80U;     /* lock PFS */
00225
00226   /* Step 3: PMR=1 to route P20/P21 to RIIC1. */
00227   REG8(0x0008C062U) |= (uint8_t)((1U << 0) | (1U << 1));
00228
00229   /* Step 4: per production riic.c: pulse internal reset (IICRST=1 then
00230    * IICRST=0) so the RIIC enters configuration state with both IICRST
00231    * and ICEEN cleared. Configuration registers are writable in this
00232    * state. Setting ICBRH/ICBRL while IICRST=1 silently no-ops -- we
00233    * already saw that empirically. */
00234   *r1(k_riic_off_iccr1) = k_iccr1_iicrst;       /* assert internal reset */
00235   *r1(k_riic_off_iccr1) = 0x00U;                /* clear all bits (IICRST=0) */
00236
00237   /* Step 5: bit rate. RX72N HW manual Table 38.x: lower 5 bits hold
00238    * the BRL/BRH count; upper 3 bits are reserved (read-as-1, write
00239    * value ignored). Writing the raw 5-bit value is fine. Production
00240    * uses internal_calculate_bit_rate(); we hard-code values that give
00241    * ~100 kHz at PCLKB=60 MHz (or scaled if PCLKB is actually 120). */
00242   *r1(k_riic_off_icbrh) = 23U;
00243   *r1(k_riic_off_icbrl) = 27U;
00244
00245   /* Step 6: mode/timing. Mirror production riic_init: only program ICMR1
00246    * (controller-mode, 7-bit addressing); leave ICMR2/ICMR3 at reset (0x00)
00247    * and DO NOT touch ICFER -- its reset value 0x72 enables the on-chip
00248    * noise filter (NFE) and clock-sync (SCLE) which the peripheral relies
00249    * on. Overriding it with our own bitmask clobbers NFE and silently
00250    * breaks edge detection. */
00251   *r1(k_riic_off_icmr1) = 0x08U;
00252
00253   /* Step 7: enable peripheral. */
00254   *r1(k_riic_off_iccr1) = k_iccr1_iceen;        /* ICEEN=1, IICRST=0 */
00255
00256   /* Step 8: kick RIIC out of post-enable "stuck low" by clearing SOWP and
00257    * forcing SCLO/SDAO high (open-drain release), then re-locking SOWP. The
00258    * peripheral comes out of IICRST with SCLO=SDAO=0 (driving low) and only
00259    * releases on the first STOP it sees. We synthesize that release by hand. */
00260   uint8_t iccr1 = *r1(k_riic_off_iccr1);
00261   iccr1 &= (uint8_t)~(1U << 4);                 /* clear SOWP -> writable */
00262   iccr1 |= (uint8_t)((1U << 3) | (1U << 2));    /* SCLO=1, SDAO=1 (release) */
00263   *r1(k_riic_off_iccr1) = iccr1;
00264   iccr1 |= (uint8_t)(1U << 4);                  /* re-lock SOWP */
00265   *r1(k_riic_off_iccr1) = iccr1;
00266 }
00267
00268 /* Force-reset RIIC1 internal state machine (clears stuck BBSY / lingering
00269  * START / NACKF flags from a botched transaction). Cheaper than full
00270  * riic1_init() because we keep PFS / MSTPCR / bit-rate untouched. */
00271 static void riic1_recover(void)
00272 {
00273   *r1(k_riic_off_iccr1) = k_iccr1_iicrst;       /* assert IICRST (ICEEN=0) */
00274   *r1(k_riic_off_iccr1) = 0x00U;                /* deassert, configuration state */
00275   /* IICRST cycle wipes ICBRH/L/ICMRx/ICFER back to reset (0xFF). Restore. */
00276   *r1(k_riic_off_icbrh) = 23U;
00277   *r1(k_riic_off_icbrl) = 27U;
00278   *r1(k_riic_off_icmr1) = 0x08U;
00279   *r1(k_riic_off_icmr3) = k_icmr3_ackwp;
00280   /* Drop k_icfer_tmoe -- timeout monitoring leaves RIIC stuck if SCL
00281    * doesn't toggle. Rely on our software wait_bit() bound instead. */
```

```
00282   *r1(k_riic_off_icfer) = (uint8_t)(k_icfer_male | k_icfer_nale |
00283                                     k_icfer_sale | k_icfer_nacke | k_icfer_scle);
00284   *r1(k_riic_off_icsr2) = 0x00U;                  /* clear all status flags */
00285   *r1(k_riic_off_iccr1) = k_iccr1_iceen;          /* re-enable */
00286 }
00287
00288 /* Polling helper with hard upper bound (NASA Rule 2). Returns false on
00289  * timeout instead of hanging. */
00290 typedef enum : uint32_t {
00291   k_poll_max = 100000U,                         /* ~few ms at -O1 */
00292 } poll_max_t;
00293
00294 static bool wait_bit(volatile uint8_t *reg, uint8_t mask, bool want_set)
00295 {
00296   for (uint32_t i = 0; i < k_poll_max; i++) {
00297     bool now = ((*reg) & mask) != 0U;
00298     if (now == want_set) { return true; }
00299   }
00300   return false;
00301 }
00302
00303 /* Per-gate failure tag, set by i2c_read_byte before a goto stop_fail jump,
00304  * so the caller can print exactly which step bailed out. */
00305 static volatile uint8_t s_last_fail_gate = 0U;
00306
00307 /* Single-byte register read: START -> addr|W -> reg -> RS -> addr|R -> data
00308  * -> NACK -> STOP.  Returns false (and forces a STOP) on any NACK/timeout. */
00309 static bool i2c_read_byte(uint8_t dev_addr, uint8_t reg_addr, uint8_t *out)
00310 {
00311   volatile uint8_t *iccr2 = r1(k_riic_off_iccr2);
00312   volatile uint8_t *icsr2 = r1(k_riic_off_icsr2);
00313   volatile uint8_t *icdrt = r1(k_riic_off_icdrt);
00314   volatile uint8_t *icdrr = r1(k_riic_off_icdrr);
00315   volatile uint8_t *icmr3 = r1(k_riic_off_icmr3);
00316
00317   if (!wait_bit(iccr2, k_iccr2_bbsy, false)) {
00318     /* Stuck busy from a previous botched transaction -- nuke the FSM. */
00319     riic1_recover();
00320     if (!wait_bit(iccr2, k_iccr2_bbsy, false)) { return false; }
00321   }
00322   /* Pre-clear any lingering status from the previous read. */
00323   *icsr2 = 0x00U;
00324
00325   /* Controller transmit mode -- mirrors production internal_riic_write_phase().
00326    * Writing MST|TRS while bus is idle is permitted; the bits become read-only
00327    * status during an active transaction. */
00328   *iccr2 = (uint8_t)(k_iccr2_mst | k_iccr2_trs);
00329   *iccr2 |= k_iccr2_st;
00330   if (!wait_bit(icsr2, k_icsr2_start, true)) { s_last_fail_gate = 1U; goto stop_fail; }
00331   *icsr2 &= (uint8_t)~k_icsr2_start;
00332
00333   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_fail_gate = 2U; goto stop_fail; }
00334   *icdrt = (uint8_t)((dev_addr « 1) | 0U);
00335   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_fail_gate = 3U; goto stop_fail; }
00336   if ((*icsr2) & k_icsr2_nackf)             { s_last_fail_gate = 4U; goto stop_fail; }
00337
00338   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_fail_gate = 5U; goto stop_fail; }
00339   *icdrt = reg_addr;
00340   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_fail_gate = 6U; goto stop_fail; }
00341   if ((*icsr2) & k_icsr2_nackf)             { s_last_fail_gate = 7U; goto stop_fail; }
00342
00343   /* Repeated start, then switch to receive mode (TRS=0, MST stays). */
00344   *iccr2 |= k_iccr2_rs;
00345   if (!wait_bit(icsr2, k_icsr2_start, true)) { s_last_fail_gate = 8U; goto stop_fail; }
00346   *icsr2 &= (uint8_t)~k_icsr2_start;
00347   *iccr2 = k_iccr2_mst;
00348
00349   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_fail_gate = 9U; goto stop_fail; }
00350   *icdrt = (uint8_t)((dev_addr « 1) | 1U);
00351   /* Receive-mode address: skip TEND. After ACK, RIIC immediately clocks the
00352    * first data byte and sets RDRF -- TEND never asserts for the addr|R byte
00353    * in controller-receive mode. NACK still surfaces via NACKF if the
00354    * peripheral didn't answer, so we check that before waiting for RDRF. */
00355   if (!wait_bit(icsr2, (uint8_t)(k_icsr2_rdrf | k_icsr2_nackf), true)) {
00356     s_last_fail_gate = 10U;
00357     goto stop_fail;
00358   }
00359   if ((*icsr2) & k_icsr2_nackf)             { s_last_fail_gate = 11U; goto stop_fail; }
00360
00361   *icmr3 |= k_icmr3_ackbt;                      /* NACK after the byte we want */
00362   (void)*icdrr;                                 /* dummy read kicks reception */
00363
00364   if (!wait_bit(icsr2, k_icsr2_rdrf, true)) { s_last_fail_gate = 12U; goto stop_fail; }
00365
00366   *icsr2 &= (uint8_t)~k_icsr2_stop;
00367   *iccr2 |= k_iccr2_sp;
00368   *out = *icdrr;
```

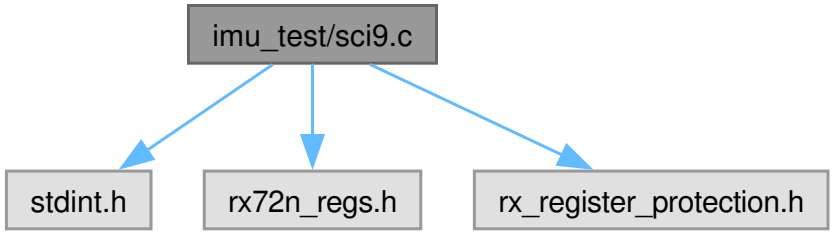
```
00369
00370   if (!wait_bit(icsr2, k_icsr2_stop, true)) { goto cleanup_only; }
00371 cleanup_only:
00372   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00373   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00374   return (*out != 0xFFU || true);               /* return true regardless of value */
00375
00376 stop_fail:
00377   *iccr2 |= k_iccr2_sp;
00378   (void)wait_bit(icsr2, k_icsr2_stop, true);
00379   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00380   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00381   return false;
00382 }
00383
00384 /* Per-write gate trace, populated by i2c_write_byte before goto wstop_fail. */
00385 static volatile uint8_t s_last_write_gate = 0U;
00386
00387 /* Single-byte register write: START -> addr|W -> reg -> val -> STOP. */
00388 static bool i2c_write_byte(uint8_t dev_addr, uint8_t reg_addr, uint8_t val)
00389 {
00390   volatile uint8_t *iccr2 = r1(k_riic_off_iccr2);
00391   volatile uint8_t *icsr2 = r1(k_riic_off_icsr2);
00392   volatile uint8_t *icdrt = r1(k_riic_off_icdrt);
00393
00394   if (!wait_bit(iccr2, k_iccr2_bbsy, false)) {
00395     riic1_recover();
00396     if (!wait_bit(iccr2, k_iccr2_bbsy, false)) { s_last_write_gate = 1U; return false; }
00397   }
00398   *icsr2 = 0x00U;
00399
00400   *iccr2 = (uint8_t)(k_iccr2_mst | k_iccr2_trs);
00401   *iccr2 |= k_iccr2_st;
00402   if (!wait_bit(icsr2, k_icsr2_start, true)) { s_last_write_gate = 2U; goto wstop_fail; }
00403   *icsr2 &= (uint8_t)~k_icsr2_start;
00404
00405   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_write_gate = 3U; goto wstop_fail; }
00406   *icdrt = (uint8_t)((dev_addr « 1) | 0U);
00407   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_write_gate = 4U; goto wstop_fail; }
00408   if ((*icsr2) & k_icsr2_nackf)             { s_last_write_gate = 5U; goto wstop_fail; }
00409
00410   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_write_gate = 6U; goto wstop_fail; }
00411   *icdrt = reg_addr;
00412   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_write_gate = 7U; goto wstop_fail; }
00413   if ((*icsr2) & k_icsr2_nackf)             { s_last_write_gate = 8U; goto wstop_fail; }
00414
00415   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { s_last_write_gate = 9U; goto wstop_fail; }
00416   *icdrt = val;
00417   if (!wait_bit(icsr2, k_icsr2_tend, true)) { s_last_write_gate = 10U; goto wstop_fail; }
00418   if ((*icsr2) & k_icsr2_nackf)             { s_last_write_gate = 11U; goto wstop_fail; }
00419
00420   *icsr2 &= (uint8_t)~k_icsr2_stop;
00421   *iccr2 |= k_iccr2_sp;
00422   (void)wait_bit(icsr2, k_icsr2_stop, true);
00423   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00424   return true;
00425
00426 wstop_fail:
00427   *iccr2 |= k_iccr2_sp;
00428   (void)wait_bit(icsr2, k_icsr2_stop, true);
00429   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00430   return false;
00431 }
00432
00433 /* Multi-byte sequential read: START -> addr|W -> reg -> RS -> addr|R ->
00434  * read N bytes (ACK first N-1, NACK last) -> STOP. Used for BNO055 sensor
00435  * burst reads (gravity vector = 6 bytes). */
00436 static bool i2c_read_bytes(uint8_t dev_addr, uint8_t reg_addr, uint8_t *buf, uint8_t count)
00437 {
00438   if (count == 0U) { return false; }
00439
00440   volatile uint8_t *iccr2 = r1(k_riic_off_iccr2);
00441   volatile uint8_t *icsr2 = r1(k_riic_off_icsr2);
00442   volatile uint8_t *icdrt = r1(k_riic_off_icdrt);
00443   volatile uint8_t *icdrr = r1(k_riic_off_icdrr);
00444   volatile uint8_t *icmr3 = r1(k_riic_off_icmr3);
00445
00446   if (!wait_bit(iccr2, k_iccr2_bbsy, false)) {
00447     riic1_recover();
00448     if (!wait_bit(iccr2, k_iccr2_bbsy, false)) { return false; }
00449   }
00450   *icsr2 = 0x00U;
00451
00452   *iccr2 = (uint8_t)(k_iccr2_mst | k_iccr2_trs);
00453   *iccr2 |= k_iccr2_st;
00454   if (!wait_bit(icsr2, k_icsr2_start, true)) { goto rstop_fail; }
00455   *icsr2 &= (uint8_t)~k_icsr2_start;
```

```
00456
00457   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { goto rstop_fail; }
00458   *icdrt = (uint8_t)((dev_addr « 1) | 0U);
00459   if (!wait_bit(icsr2, k_icsr2_tend, true)) { goto rstop_fail; }
00460   if ((*icsr2) & k_icsr2_nackf)             { goto rstop_fail; }
00461
00462   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { goto rstop_fail; }
00463   *icdrt = reg_addr;
00464   if (!wait_bit(icsr2, k_icsr2_tend, true)) { goto rstop_fail; }
00465   if ((*icsr2) & k_icsr2_nackf)             { goto rstop_fail; }
00466
00467   *iccr2 |= k_iccr2_rs;
00468   if (!wait_bit(icsr2, k_icsr2_start, true)) { goto rstop_fail; }
00469   *icsr2 &= (uint8_t)~k_icsr2_start;
00470   *iccr2 = k_iccr2_mst;
00471
00472   if (!wait_bit(icsr2, k_icsr2_tdre, true)) { goto rstop_fail; }
00473   *icdrt = (uint8_t)((dev_addr « 1) | 1U);
00474   if (!wait_bit(icsr2, (uint8_t)(k_icsr2_rdrf | k_icsr2_nackf), true)) { goto rstop_fail; }
00475   if ((*icsr2) & k_icsr2_nackf)             { goto rstop_fail; }
00476
00477   /* Dummy read ICDRR to release SCL and start clocking byte 0 into the
00478    * shift register. RX72N HW manual section 38.2.5.3 (controller-receive). */
00479   if (count == 1U) {
00480     /* For a single-byte read, pre-set ACKBT=1 BEFORE the dummy read so the
00481      * byte's ACK slot sends NACK, signalling the peripheral to stop. */
00482     *icmr3 |= k_icmr3_ackbt;
00483   }
00484   (void)*icdrr;
00485
00486   for (uint8_t i = 0; i < count; i++) {
00487     if (!wait_bit(icsr2, k_icsr2_rdrf, true)) { goto rstop_fail; }
00488     if (i == (uint8_t)(count - 2U)) {
00489       /* Next byte is the final one -- set ACKBT=1 so the byte we're about
00490        * to read carries NACK in its 9th-bit slot. */
00491       *icmr3 |= k_icmr3_ackbt;
00492     }
00493     if (i == (uint8_t)(count - 1U)) {
00494       *icsr2 &= (uint8_t)~k_icsr2_stop;
00495       *iccr2 |= k_iccr2_sp;
00496     }
00497     buf[i] = *icdrr;
00498   }
00499
00500   (void)wait_bit(icsr2, k_icsr2_stop, true);
00501   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00502   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00503   return true;
00504
00505 rstop_fail:
00506   *iccr2 |= k_iccr2_sp;
00507   (void)wait_bit(icsr2, k_icsr2_stop, true);
00508   *icsr2 &= (uint8_t)~(k_icsr2_stop | k_icsr2_nackf);
00509   *icmr3 &= (uint8_t)~k_icmr3_ackbt;
00510   return false;
00511 }
00512
00513 /* ---------------------------------------------------------------------- */
00514 typedef enum : uint8_t {
00515   k_bno055_addr        = 0x28U,
00516   k_bno055_chip_id_reg = 0x00U,
00517   k_bno055_acc_x_lsb   = 0x08U,
00518   k_bno055_grv_x_lsb   = 0x2EU,  /* Gravity vector base, 6 bytes int16 LE, 1 LSB = 1/100 m/s^2 */
00519   k_bno055_lia_x_lsb   = 0x28U,  /* Linear acceleration base (gravity removed by fusion) */
00520   k_bno055_opr_mode    = 0x3DU,  /* Operation mode register */
00521   k_bno055_pwr_mode    = 0x3EU,  /* Power mode register */
00522   k_bno055_sys_trigger = 0x3FU,  /* System trigger (incl. self-test, soft reset) */
00523   k_bno055_unit_sel    = 0x3BU,  /* Output units selection */
00524   k_bno055_page_id     = 0x07U,  /* Register page selector */
00525   k_bno055_calib_stat  = 0x35U,  /* Calibration status (SYS|GYR|ACC|MAG, 2 bits each) */
00526   k_bno055_mode_config = 0x00U,  /* OPR_MODE: CONFIGMODE */
00527   k_bno055_mode_ndof   = 0x0CU,  /* OPR_MODE: NDOF (full sensor fusion) */
00528   k_bno055_pwr_normal  = 0x00U,  /* PWR_MODE: normal */
00529   k_bmp280_addr        = 0x76U,
00530   k_bmp280_chip_id_reg = 0xD0U,
00531 } sensor_t;
00532
00533 int main(void)
00534 {
00535   clock_init();
00536   sci9_debug_init();
00537   /* 5-second window so the operator can attach `cat /dev/ttyACM0' after a
00538    * fresh flash-and-reset and still see step 1..4 messages. */
00539   busy_wait_ms(5000);
00540   sci9_debug_puts("\n=== IMU_TEST start ===\n");
00541
00542   sci9_debug_puts("step 1: imu_reset_pulse (P83 hi-lo-hi + 700 ms)\n");
```

```
00543  imu_reset_pulse();
00544  sci9_debug_puts("  done\n");
00545
00546  sci9_debug_puts("step 2: riic1_init\n");
00547  riic1_init();
00548  sci9_debug_puts("  done\n");
00549
00550  sci9_debug_puts("post-init regs:");
00551  sci9_debug_puts(" ICCR1=0x"); sci9_debug_puthex16((uint16_t)*r1(k_riic_off_iccr1));
00552  sci9_debug_puts(" ICCR2=0x"); sci9_debug_puthex16((uint16_t)*r1(k_riic_off_iccr2));
00553  sci9_debug_puts(" ICSR2=0x"); sci9_debug_puthex16((uint16_t)*r1(k_riic_off_icsr2));
00554  sci9_debug_puts("\n");
00555
00556  sci9_debug_puts("step 3: probe BNO055 chip_id @ 0x28 reg 0x00\n");
00557  uint8_t bno_id = 0xFFU;
00558  bool bno_ok = i2c_read_byte(k_bno055_addr, k_bno055_chip_id_reg, &bno_id);
00559  sci9_debug_puts(bno_ok ? "  read ok, chip_id=0x" : "  read FAILED, last=0x");
00560  sci9_debug_puthex16((uint16_t)bno_id);
00561  sci9_debug_puts(bno_id == 0xA0U ? " (BNO055 ok)\n" : " (expected 0xA0)\n");
00562
00563  sci9_debug_puts("step 4: probe BMP280 chip_id @ 0x76 reg 0xD0\n");
00564  uint8_t bmp_id = 0xFFU;
00565  bool bmp_ok = i2c_read_byte(k_bmp280_addr, k_bmp280_chip_id_reg, &bmp_id);
00566  sci9_debug_puts(bmp_ok ? "  read ok, chip_id=0x" : "  read FAILED, last=0x");
00567  sci9_debug_puthex16((uint16_t)bmp_id);
00568  sci9_debug_puts(bmp_id == 0x58U ? " (BMP280 ok)\n" : " (expected 0x58)\n");
00569
00570  sci9_debug_puts("step 5: BNO055 -> NDOF mode (re-init each xact)\n");
00571  riic1_init();
00572  s_last_write_gate = 0U;
00573  bool ok_cfg = i2c_write_byte(k_bno055_addr, k_bno055_opr_mode, k_bno055_mode_config);
00574  uint8_t cfg_gate = s_last_write_gate;
00575  busy_wait_ms(25);
00576  riic1_init();
00577  s_last_write_gate = 0U;
00578  bool ok_ndof = i2c_write_byte(k_bno055_addr, k_bno055_opr_mode, k_bno055_mode_ndof);
00579  uint8_t ndof_gate = s_last_write_gate;
00580  busy_wait_ms(25);
00581  sci9_debug_puts("  cfg="); sci9_debug_puts(ok_cfg ? "ok" : "FAIL");
00582  sci9_debug_puts("@gate=0x"); sci9_debug_puthex16((uint16_t)cfg_gate);
00583  sci9_debug_puts("  ndof="); sci9_debug_puts(ok_ndof ? "ok" : "FAIL");
00584  sci9_debug_puts("@gate=0x"); sci9_debug_puthex16((uint16_t)ndof_gate);
00585  sci9_debug_puts("\n");
00586
00587  sci9_debug_puts("step 6: gravity loop @ 5 Hz (expect |g| ~= 981)\n");
00588  sci9_debug_puts("\n------------------------------------------------------------\n");
00589  sci9_debug_puts("Watching BNO055 @ 2 Hz. All values are raw int16 from the chip.\n");
00590  sci9_debug_puts("Scale factors (per BNO055 datasheet, units = m/s^2, deg/s, deg, uT):\n");
00591  sci9_debug_puts("  ACC / LIA / GRV: divide by 100  (1 LSB = 0.01 m/s^2)\n");
00592  sci9_debug_puts("  GYR            : divide by 16   (1 LSB = 0.0625 deg/s)\n");
00593  sci9_debug_puts("  EUL (hdg/r/p)  : divide by 16   (1 LSB = 0.0625 deg)\n");
00594  sci9_debug_puts("  QUA            : divide by 16384 (1 LSB = 1 / 2^14)\n");
00595  sci9_debug_puts("  MAG            : divide by 16   (1 LSB = 0.0625 uT)\n");
00596  sci9_debug_puts("  temp           : deg C (int8 as-is)\n");
00597  sci9_debug_puts("  cal = SYS«6 | GYR«4 | ACC«2 | MAG (each 0..3)\n");
00598  sci9_debug_puts("------------------------------------------------------------\n");
00599
00600  uint16_t iter = 0;
00601  for (;;) {
00602    /* One burst covers ACC..CALIB_STAT = 46 bytes starting at 0x08. */
00603    uint8_t block[46] = {0};
00604    riic1_init();
00605    bool ok = i2c_read_bytes(k_bno055_addr, k_bno055_acc_x_lsb, block, sizeof(block));
00606
00607    if (!ok) {
00608      sci9_debug_puts("iter=");
00609      sci9_debug_puthex16(iter);
00610      sci9_debug_puts("  READ_FAIL\n");
00611      busy_wait_ms(500);
00612      iter++;
00613      continue;
00614    }
00615
00616    /* Indexing relative to start-register 0x08:
00617     *   ACC  : 0x08..0x0D -> block[ 0.. 5]
00618     *   MAG  : 0x0E..0x13 -> block[ 6..11]
00619     *   GYR  : 0x14..0x19 -> block[12..17]
00620     *   EUL  : 0x1A..0x1F -> block[18..23]  (heading, roll, pitch)
00621     *   QUA  : 0x20..0x27 -> block[24..31]  (w, x, y, z)
00622     *   LIA  : 0x28..0x2D -> block[32..37]
00623     *   GRV  : 0x2E..0x33 -> block[38..43]
00624     *   TEMP : 0x34       -> block[44]      (int8)
00625     *   CAL  : 0x35       -> block[45]      (status byte)
00626     */
00627 #define I16_LE(p, i) \
00628   ((int16_t)((uint16_t)(p)[(i)] | ((uint16_t)(p)[(i) + 1U] « 8)))
00629
```

```
00630     int16_t acc_x = I16_LE(block, 0);  int16_t acc_y = I16_LE(block, 2);  int16_t acc_z = I16_LE(block, 4);
00631     int16_t mag_x = I16_LE(block, 6);  int16_t mag_y = I16_LE(block, 8);  int16_t mag_z = I16_LE(block, 10);
00632     int16_t gyr_x = I16_LE(block, 12); int16_t gyr_y = I16_LE(block, 14); int16_t gyr_z = I16_LE(block, 16);
00633     int16_t eul_h = I16_LE(block, 18); int16_t eul_r = I16_LE(block, 20); int16_t eul_p = I16_LE(block, 22);
00634     int16_t qua_w = I16_LE(block, 24); int16_t qua_x = I16_LE(block, 26);
00635     int16_t qua_y = I16_LE(block, 28); int16_t qua_z = I16_LE(block, 30);
00636     int16_t lia_x = I16_LE(block, 32); int16_t lia_y = I16_LE(block, 34); int16_t lia_z = I16_LE(block, 36);
00637     int16_t grv_x = I16_LE(block, 38); int16_t grv_y = I16_LE(block, 40); int16_t grv_z = I16_LE(block, 42);
00638     int8_t  temp  = (int8_t)block[44];
00639     uint8_t calib = block[45];
00640
00641 #undef I16_LE
00642
00643     sci9_debug_puts("iter=");
00644     sci9_debug_puthex16(iter);
00645     sci9_debug_puts("\n");
00646
00647     sci9_debug_puts("  ACC  x="); sci9_debug_putint16_col(acc_x);
00648     sci9_debug_puts("  y=");       sci9_debug_putint16_col(acc_y);
00649     sci9_debug_puts("  z=");       sci9_debug_putint16_col(acc_z);
00650     sci9_debug_puts("   (raw accel, /100 -> m/s^2)\n");
00651
00652     sci9_debug_puts("  MAG  x="); sci9_debug_putint16_col(mag_x);
00653     sci9_debug_puts("  y=");       sci9_debug_putint16_col(mag_y);
00654     sci9_debug_puts("  z=");       sci9_debug_putint16_col(mag_z);
00655     sci9_debug_puts("   (magnetometer, /16 -> uT)\n");
00656
00657     sci9_debug_puts("  GYR  x="); sci9_debug_putint16_col(gyr_x);
00658     sci9_debug_puts("  y=");       sci9_debug_putint16_col(gyr_y);
00659     sci9_debug_puts("  z=");       sci9_debug_putint16_col(gyr_z);
00660     sci9_debug_puts("   (gyroscope, /16 -> deg/s)\n");
00661
00662     sci9_debug_puts("  EUL  h="); sci9_debug_putint16_col(eul_h);
00663     sci9_debug_puts("  r=");       sci9_debug_putint16_col(eul_r);
00664     sci9_debug_puts("  p=");       sci9_debug_putint16_col(eul_p);
00665     sci9_debug_puts("   (euler heading/roll/pitch, /16 -> deg)\n");
00666
00667     sci9_debug_puts("  QUA  w="); sci9_debug_putint16_col(qua_w);
00668     sci9_debug_puts("  x=");       sci9_debug_putint16_col(qua_x);
00669     sci9_debug_puts("  y=");       sci9_debug_putint16_col(qua_y);
00670     sci9_debug_puts("  z=");       sci9_debug_putint16_col(qua_z);
00671     sci9_debug_puts("  (quaternion, /16384)\n");
00672
00673     sci9_debug_puts("  LIA  x="); sci9_debug_putint16_col(lia_x);
00674     sci9_debug_puts("  y=");       sci9_debug_putint16_col(lia_y);
00675     sci9_debug_puts("  z=");       sci9_debug_putint16_col(lia_z);
00676     sci9_debug_puts("   (linear accel, gravity removed, /100)\n");
00677
00678     sci9_debug_puts("  GRV  x="); sci9_debug_putint16_col(grv_x);
00679     sci9_debug_puts("  y=");       sci9_debug_putint16_col(grv_y);
00680     sci9_debug_puts("  z=");       sci9_debug_putint16_col(grv_z);
00681     sci9_debug_puts("   (gravity vector, /100)\n");
00682
00683     sci9_debug_puts("  temp="); sci9_debug_putint16_col((int16_t)temp);
00684     sci9_debug_puts(" degC     cal=0x"); sci9_debug_puthex16((uint16_t)calib);
00685     sci9_debug_puts(" (SYS="); sci9_debug_putint16_col((int16_t)((calib » 6) & 0x3U));
00686     sci9_debug_puts(" GYR="); sci9_debug_putint16_col((int16_t)((calib » 4) & 0x3U));
00687     sci9_debug_puts(" ACC="); sci9_debug_putint16_col((int16_t)((calib » 2) & 0x3U));
00688     sci9_debug_puts(" MAG="); sci9_debug_putint16_col((int16_t)(calib & 0x3U));
00689     sci9_debug_puts(")\n\n");
00690
00691     busy_wait_ms(500);
00692     iter++;
00693   }
00694 }
```

```
#include <stdint.h>
  #include "rx72n_regs.h"
  #include "rx_register_protection.h"
```

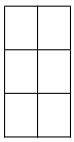


<<<<

*
*
*
*
*
*
*

*

enum hex_shifts_t : uint8_t

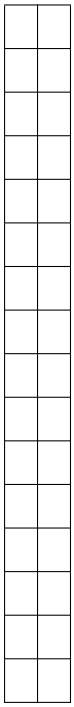


```
00062             : uint8_t {
00063     k_hex_shift_init32 = 28U,
00064     k_hex_shift_init16 = 12U,
00065     k_hex_shift_step   = 4U,
00066 } hex_shifts_t;
```

enum sci9_addrs_t : uintptr_t

```
00035             : uintptr_t {
00036     k_sci9_base = 0x000D0020U,
00037 } sci9_addrs_t;
```

enum sci9_cfg_t : uint8_t



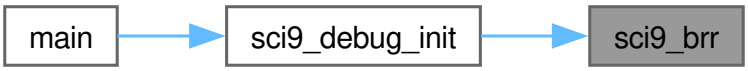
```
00039             : uint8_t {
00040     k_psel_sci9        = 0x0AU, /**< SCI9 RXD/TXD function */
00041     k_pb6_mask          = (uint8_t)(1U << 6U),
00042     k_pb7_mask          = (uint8_t)(1U << 7U),
00043     k_mstpc_sci9_bit    = 26U, /* MSTPCRC.MSTPC26 per RX72N HW manual page 410 */
00044     k_pwpr_unlock_b0    = 0x00U,
00045     k_pwpr_pfswe        = 0x40U,
00046     k_pwpr_b0wi         = 0x80U,
00047     k_brr         = 31U, /* SCI9 uses PCLKA=120 MHz (SCI7-11 group); BRR=31 @ CKS=0 ABCS=0 -> ~117 kbaud
        (1.7% over 115200) */
00048     k_smr_n1_async      = 0x00U,
00049     k_scr_te_only       = 0x20U, /**< CKE=00, MPIE=0, RIE=0, TE=1, RE=0 */
00050     k_scr_txrx_enabled = 0x30U, /**< CKE=00, MPIE=0, RIE=0, TE=1, RE=1 */
00051     k_scr_disabled      = 0x00U,
00052     k_ssr_tdre          = 0x80U,
00053     k_scmr_default      = 0xF2U,
00054     k_semr_default      = 0x00U,
00055     k_hex_nibble_mask   = 0x0FU,
00056 } sci9_cfg_t;
```

enum sci9_settle_t : uint16_t

```
00058             : uint16_t {
00059     k_brr_settle_loops = 1000U, /**< Match production uart.c (~8.68 us @ 115200) */
00060 } sci9_settle_t;
```

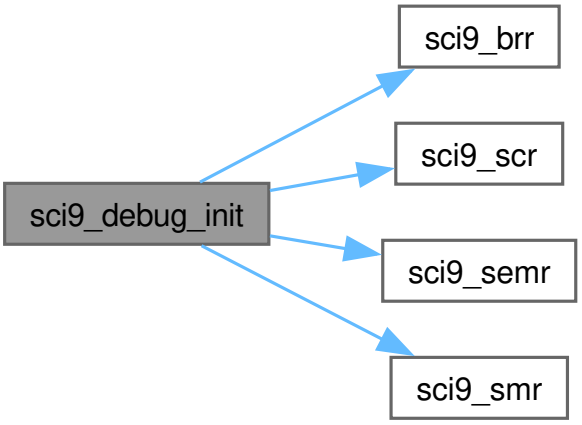
volatile uint8_t * sci9_brr (
void) [inline], [static]

```
00077 {
00078     return (volatile uint8_t *)(k_sci9_base + 1U);
00079 }
```



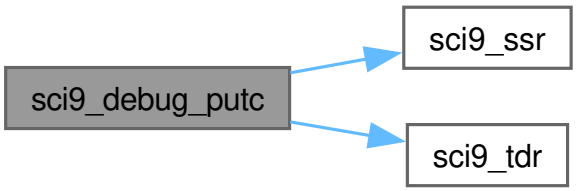

```
void sci9_debug_init (
                    void )
```

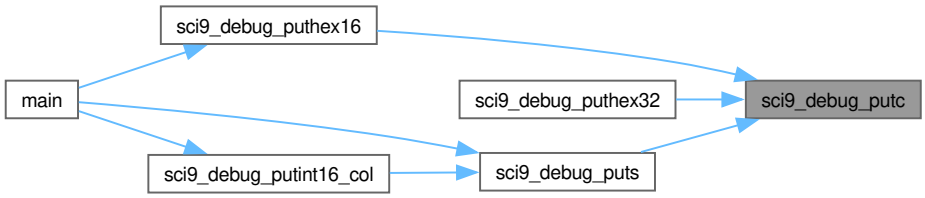
```
00105 {
00106     /* Step 1: release SCI9 from module stop. SCI9 = MSTPCRC bit 26 per
00107      * RX72N HW manual page 410 (NOT MSTPCRB.bit22 -- that's PDC). */
00108     *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00109     system_regs()->mstpcrc &= ~(uint32_t)(1UL « (uint32_t)k_mstpc_sci9_bit);
00110     *prcr_reg() = k_rx_prcr_lock;
00111
00112     /* Step 2: configure PB6 (RXD9) and PB7 (TXD9) via MPC.  Use the struct
00113      * accessor so the compiler computes the correct PFS offsets via
00114      * static_assert-checked layout. */
00115     mpc()->pwpr   = k_pwpr_unlock_b0; /* clear B0WI */
00116     mpc()->pwpr   = k_pwpr_pfswe;     /* allow PFS writes */
00117     mpc()->pb6pfs = k_psel_sci9;
00118     mpc()->pb7pfs = k_psel_sci9;
00119     mpc()->pwpr   = k_pwpr_unlock_b0; /* clear PFSWE */
00120     mpc()->pwpr   = k_pwpr_b0wi;      /* re-protect */
00121
00122     /* Step 3: PDR before PMR (production uart.c does this).  PB7 = output
00123      * for TX, PB6 = input for RX.  Then flip PMR to peripheral mode. */
00124     portb()->pdr |= (uint8_t)k_pb7_mask;
00125     portb()->pdr &= (uint8_t) ~k_pb6_mask;
00126     portb()->pmr |= (uint8_t)(k_pb6_mask | k_pb7_mask);
00127
00128     /* Step 4: SCI9 init.  Match the production order:
00129      *   SCR=0 -> SMR -> BRR -> settle -> SCR=TE|RE -> SEMR.
00130      * Don't touch SCMR (reset default 0xF2 is correct). */
00131     *sci9_scr() = k_scr_disabled;
00132     *sci9_smr() = k_smr n1_async;
00133     *sci9_brr() = k_brr ;
00134
00135     /* Wait roughly one bit time before enabling TX/RX. */
00136     for (volatile uint32_t i = 0U; i < (uint32_t)k_brr_settle_loops; i++) {
00137         __asm__ volatile("nop");
00138     }
00139
00140     *sci9_scr()  = k_scr_txrx_enabled;
00141     *sci9_semr() = k_semr_default;
00142 }
```



void sci9_debug_putc (
char c)

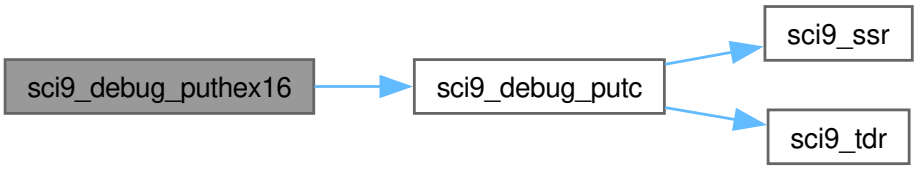
```
00148 {
00149     while ((*sci9_ssr() & (uint8_t)k_ssr_tdre) == 0U) {
00150         /* spin */
00151     }
00152     *sci9_tdr() = (uint8_t)c;
00153 }
```

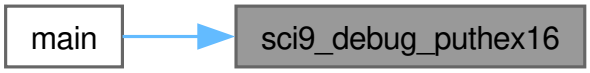




void sci9_debug_puthex16 (
uint16_t v)

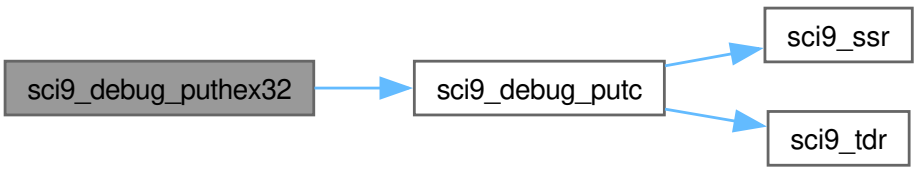
```
00180 {
00181     static const char hex_digits[] = "0123456789ABCDEF";
00182     sci9_debug_putc('0');
00183     sci9_debug_putc('x');
00184     for (int8_t shift = (int8_t)k_hex_shift_init16; shift >= 0;
00185          shift -= (int8_t)k_hex_shift_step) {
00186         sci9_debug_putc(hex_digits[(v » (uint8_t)shift) & (uint8_t)k_hex_nibble_mask]);
00187     }
00188 }
```





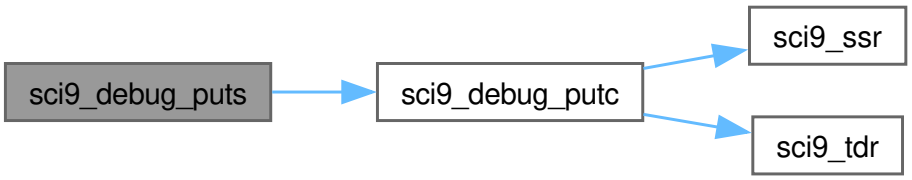
void sci9_debug_puthex32 (
uint32_t v)

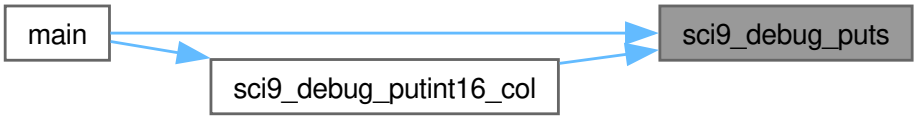
```
00169 {
00170     static const char hex_digits[] = "0123456789ABCDEF";
00171     sci9_debug_putc('0');
00172     sci9_debug_putc('x');
00173     for (int8_t shift = (int8_t)k_hex_shift_init32; shift >= 0;
00174          shift -= (int8_t)k_hex_shift_step) {
00175         sci9_debug_putc(hex_digits[(v » (uint8_t)shift) & (uint8_t)k_hex_nibble_mask]);
00176     }
00177 }
```



void sci9_debug_puts (
const char * s)

```
00156 {
00157     if (s == (const char *)0) {
00158         return;
00159     }
00160     while (*s != '\0') {
00161         if (*s == '\n') {
00162             sci9_debug_putc('\r');
00163         }
00164         sci9_debug_putc(*s++);
00165     }
00166 }
```



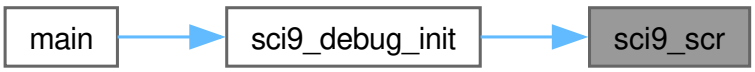


volatile uint8_t * sci9_scmr (void) [inline], [static]

```
00093 {
00094     return (volatile uint8_t *)(k_sci9_base + 6U);
00095 }
```

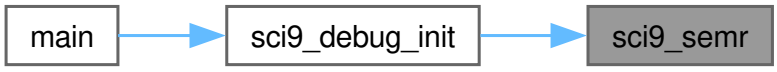
volatile uint8_t * sci9_scr (void) [inline], [static]

```
00081 {
00082     return (volatile uint8_t *)(k_sci9_base + 2U);
00083 }
```



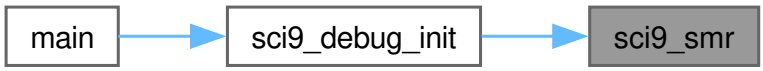
volatile uint8_t * sci9_semr (void) [inline], [static]

```
00097 {
00098     return (volatile uint8_t *)(k_sci9_base + 7U);
00099 }
```



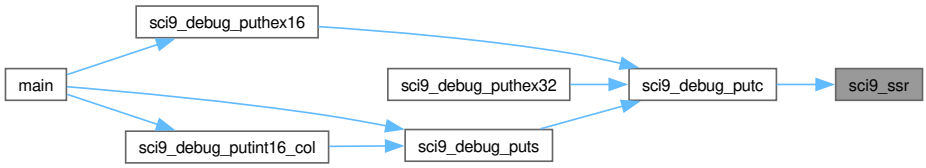
volatile uint8_t * sci9_smr (
void) [inline], [static]

```
00073 {
00074     return (volatile uint8_t *)(k_sci9_base + 0U);
00075 }
```



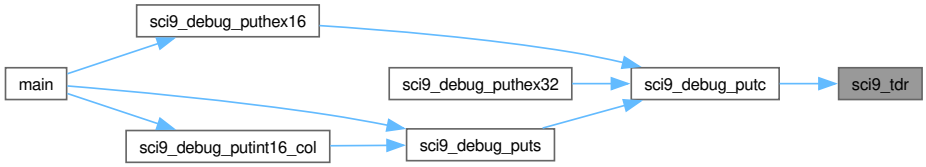
volatile uint8_t * sci9_ssr (
void) [inline], [static]

```
00089 {
00090     return (volatile uint8_t *)(k_sci9_base + 4U);
00091 }
```



volatile uint8_t * sci9_tdr (
void) [inline], [static]

```
00085 {
00086     return (volatile uint8_t *)(k_sci9_base + 3U);
00087 }
```

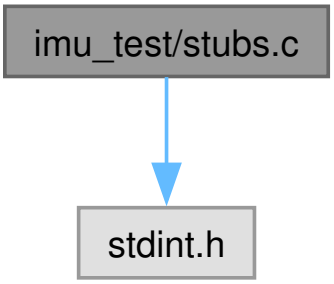


```
00001 /**
00002  * @file sci9_debug.c
00003  * @brief Minimal polled-mode SCI9 (TXD9 = PB7, RXD9 = PB6) debug UART for imu_test.
00004  *
00005  * @details
00006  * The STAR board routes SCI9 to the on-board CY7C65213 USB-to-UART bridge
00007  * which appears as /dev/ttyACM0 on the host (or COMx on Windows).  This
00008  * file provides just enough init + putc/puts to emit ASCII status from
00009  * the firmware so we can debug the USB CDC bring-up without needing the
00010  * USB device to actually enumerate.
00011  *
00012  * Uses the rx_hal struct accessors (mpc(), portb(), system_regs(), prcr_reg())
00013  * so the addresses are all compiler-verified via static_assert.
00014  *
00015  * Baud = 115200. SCI9 is in the SCI7-11 group that uses PCLKA (120 MHz),
00016  * not PCLKB. At CKS=0, ABCS=0:
00017  *   BRR = (PCLKA / (32 * baud)) - 1 = 120000000 / 3686400 - 1 = 31.55 -> 31
00018  *   Actual baud = PCLKA / (32 * (BRR+1)) = 117187 Hz (+1.7% vs 115200, in tolerance).
00019  *
00020  * SPDX-License-Identifier: MIT
00021  * @copyright Copyright (c) 2026 Locked Inc.
00022  */
00023
00024 #include <stdint.h>
00025
00026 #include "rx72n_regs.h"
00027 #include "rx_register_protection.h"
00028
00029 /*
      ================================================================================
00030  * SCI9 register access via raw addresses (rx72n_sci_regs.h doesn't expose
00031  * the per-channel struct as cleanly as we'd like for a quick polled driver).
00032  * The base 0x000D0020 is taken from the verified header.
00033  * Register offsets per RX72N HW manual: SMR=0, BRR=1, SCR=2, TDR=3, SSR=4.
00034  *
      ================================================================================
      */
00035 typedef enum : uintptr_t {
00036     k_sci9_base = 0x000D0020U,
00037 } sci9_addrs_t;
00038
00039 typedef enum : uint8_t {
00040     k_psel_sci9        = 0x0AU, /**< SCI9 RXD/TXD function */
00041     k_pb6_mask         = (uint8_t)(1U << 6U),
00042     k_pb7_mask         = (uint8_t)(1U << 7U),
00043     k_mstpc_sci9_bit   = 26U, /* MSTPCRC.MSTPC26 per RX72N HW manual page 410 */
00044     k_pwpr_unlock_b0   = 0x00U,
00045     k_pwpr_pfswe       = 0x40U,
00046     k_pwpr_b0wi        = 0x80U,
00047     k_brr        = 31U, /* SCI9 uses PCLKA=120 MHz (SCI7-11 group); BRR=31 @ CKS=0 ABCS=0 -> ~117 kbaud
      (1.7% over 115200) */
00048     k_smr_n1_async     = 0x00U,
00049     k_scr_te_only      = 0x20U, /**< CKE=00, MPIE=0, RIE=0, TE=1, RE=0 */
00050     k_scr_txrx_enabled = 0x30U, /**< CKE=00, MPIE=0, RIE=0, TE=1, RE=1 */
00051     k_scr_disabled     = 0x00U,
00052     k_ssr_tdre         = 0x80U,
00053     k_scmr_default     = 0xF2U,
00054     k_semr_default     = 0x00U,
00055     k_hex_nibble_mask  = 0x0FU,
00056 } sci9_cfg_t;
00057
00058 typedef enum : uint16_t {
00059     k_brr_settle_loops = 1000U, /**< Match production uart.c (~8.68 us @ 115200) */
00060 } sci9_settle_t;
00061
00062 typedef enum : uint8_t {
00063     k_hex_shift_init32 = 28U,
00064     k_hex_shift_init16 = 12U,
00065     k_hex_shift_step   = 4U,
00066 } hex_shifts_t;
00067
00068 /*
      ================================================================================
00069  * SCI9 register accessors (raw addresses since the header doesn't give us a
00070  * per-channel struct for the SCIi/SCIj quirks).
00071  *
      ================================================================================
      */
00072 static inline volatile uint8_t *sci9_smr(void)
00073 {
00074     return (volatile uint8_t *)(k_sci9_base + 0U);
00075 }
00076 static inline volatile uint8_t *sci9_brr(void)
00077 {
00078     return (volatile uint8_t *)(k_sci9_base + 1U);
00079 }
```

```
00080 static inline volatile uint8_t *sci9__scr(void)
00081 {
00082     return (volatile uint8_t *)(k__sci9__base + 2U);
00083 }
00084 static inline volatile uint8_t *sci9__tdr(void)
00085 {
00086     return (volatile uint8_t *)(k__sci9__base + 3U);
00087 }
00088 static inline volatile uint8_t *sci9__ssr(void)
00089 {
00090     return (volatile uint8_t *)(k__sci9__base + 4U);
00091 }
00092 static inline volatile uint8_t *sci9__scmr(void)
00093 {
00094     return (volatile uint8_t *)(k__sci9__base + 6U);
00095 }
00096 static inline volatile uint8_t *sci9__semr(void)
00097 {
00098     return (volatile uint8_t *)(k__sci9__base + 7U);
00099 }
00100
00101 /*
    ============================================================================
00102  * Initialise SCI9 for 115200 8N1 polled output on PB7 (TXD9).
00103  *
    ============================================================================
    */
00104 void sci9__debug__init(void)
00105 {
00106     /* Step 1: release SCI9 from module stop. SCI9 = MSTPCRC bit 26 per
00107      * RX72N HW manual page 410 (NOT MSTPCRB.bit22 -- that's PDC). */
00108     *prcr__reg() = k__rx__prcr__unlock__prc0__prc1;
00109     system__regs()->mstpcrc &= ~(uint32__t)(1UL « (uint32__t)k__mstpc__sci9__bit);
00110     *prcr__reg() = k__rx__prcr__lock;
00111
00112     /* Step 2: configure PB6 (RXD9) and PB7 (TXD9) via MPC.  Use the struct
00113      * accessor so the compiler computes the correct PFS offsets via
00114      * static_assert-checked layout. */
00115     mpc()->pwpr   = k__pwpr__unlock__b0; /* clear B0WI */
00116     mpc()->pwpr   = k__pwpr__pfswe;      /* allow PFS writes */
00117     mpc()->pb6pfs = k__psel__sci9;
00118     mpc()->pb7pfs = k__psel__sci9;
00119     mpc()->pwpr   = k__pwpr__unlock__b0; /* clear PFSWE */
00120     mpc()->pwpr   = k__pwpr__b0wi;       /* re-protect */
00121
00122     /* Step 3: PDR before PMR (production uart.c does this).  PB7 = output
00123      * for TX, PB6 = input for RX.  Then flip PMR to peripheral mode. */
00124     portb()->pdr |= (uint8__t)k__pb7__mask;
00125     portb()->pdr &= (uint8__t) ~k__pb6__mask;
00126     portb()->pmr |= (uint8__t)(k__pb6__mask | k__pb7__mask);
00127
00128     /* Step 4: SCI9 init.  Match the production order:
00129      *   SCR=0 -> SMR -> BRR -> settle -> SCR=TE|RE -> SEMR.
00130      * Don't touch SCMR (reset default 0xF2 is correct). */
00131     *sci9__scr() = k__scr__disabled;
00132     *sci9__smr() = k__smr n1__async;
00133     *sci9__brr() = k__brr ;
00134
00135     /* Wait roughly one bit time before enabling TX/RX. */
00136     for (volatile uint32__t i = 0U; i < (uint32__t)k__brr__settle__loops; i++) {
00137         ___asm___ volatile("nop");
00138     }
00139
00140     *sci9__scr()  = k__scr__txrx__enabled;
00141     *sci9__semr() = k__semr__default;
00142 }
00143
00144 /*
    ============================================================================
00145  * Polled write: spin until TDRE then drop byte into TDR.
00146  *
    ============================================================================
    */
00147 void sci9__debug__putc(char c)
00148 {
00149     while ((*sci9__ssr() & (uint8__t)k__ssr__tdre) == 0U) {
00150         /* spin */
00151     }
00152     *sci9__tdr() = (uint8__t)c;
00153 }
00154
00155 void sci9__debug__puts(const char *s)
00156 {
00157     if (s == (const char *)0) {
00158         return;
00159     }
00160     while (*s != '\0') {
```

```
00161         if (*s == '\n') {
00162             sci9_debug_putc('\r');
00163         }
00164         sci9_debug_putc(*s++);
00165     }
00166 }
00167
00168 void sci9_debug_puthex32(uint32_t v)
00169 {
00170     static const char hex_digits[] = "0123456789ABCDEF";
00171     sci9_debug_putc('0');
00172     sci9_debug_putc('x');
00173     for (int8_t shift = (int8_t)k_hex_shift_init32; shift >= 0;
00174          shift -= (int8_t)k_hex_shift_step) {
00175         sci9_debug_putc(hex_digits[(v » (uint8_t)shift) & (uint8_t)k_hex_nibble_mask]);
00176     }
00177 }
00178
00179 void sci9_debug_puthex16(uint16_t v)
00180 {
00181     static const char hex_digits[] = "0123456789ABCDEF";
00182     sci9_debug_putc('0');
00183     sci9_debug_putc('x');
00184     for (int8_t shift = (int8_t)k_hex_shift_init16; shift >= 0;
00185          shift -= (int8_t)k_hex_shift_step) {
00186         sci9_debug_putc(hex_digits[(v » (uint8_t)shift) & (uint8_t)k_hex_nibble_mask]);
00187     }
00188 }
```

#include <stdint.h>



*

*

**

```
void rx_log_uart_putc (
                char c)

 00016 { (void)c; }

void rx_log_uart_puthex (
                uint32_t v,
                uint8_t d)

 00020 { (void)v; (void)d; }

void rx_log_uart_putint (
                int32_t v)

 00018 { (void)v; }

void rx_log_uart_puts (
                const char * s)

 00017 { (void)s; }

void rx_log_uart_putuint (
                uint32_t v)

 00019 { (void)v; }

void uart_debug_putc (
                char c)

 00022 { (void)c; }

void uart_debug_puthex (
                uint32_t v,
                uint8_t d)

 00026 { (void)v; (void)d; }

void uart_debug_putint (
                int32_t v)

 00024 { (void)v; }

void uart_debug_puts (
                const char * s)

 00023 { (void)s; }
```

void uart_debug_putuint (
uint32_t v)

00025 { (void)v; }

```
00001 /**
00002  * @file stubs.c
00003  * @brief Log/UART no-op stubs for the bare-metal IMU smoke test.
00004  *
00005  * @details
00006  * The bench test does not pull in the rx_log family or the framed UART
00007  * ring buffer. Anything in the dependency chain that calls
00008  * rx_log_uart_* / uart_debug_* must have something to link against.
00009  *
00010  * SPDX-License-Identifier: MIT
00011  * @copyright Copyright (c) 2026 Locked Inc.
00012  */
00013
00014 #include <stdint.h>
00015
00016 void rx_log_uart_putc(char c)              { (void)c; }
00017 void rx_log_uart_puts(const char *s)       { (void)s; }
00018 void rx_log_uart_putint(int32_t v)         { (void)v; }
00019 void rx_log_uart_putuint(uint32_t v)       { (void)v; }
00020 void rx_log_uart_puthex(uint32_t v, uint8_t d) { (void)v; (void)d; }
00021
00022 void uart_debug_putc(char c)               { (void)c; }
00023 void uart_debug_puts(const char *s)        { (void)s; }
00024 void uart_debug_putint(int32_t v)          { (void)v; }
00025 void uart_debug_putuint(uint32_t v)        { (void)v; }
00026 void uart_debug_puthex(uint32_t v, uint8_t d) { (void)v; (void)d; }
```